

# **Banking under Conflict: Managers and Organizational Design**

## *Appendix*

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## A Model details and full derivations

### A.1 Environment, timing, and interpretation

The framework combines standard information–control models (Philippe Aghion and Jean Tirole, 1997; Ricardo Alonso, Wouter Dessein and Niko Matouschek, 2008; Wouter Dessein, 2002) with work on hierarchy and soft information in banking (Jose M. Liberti and Atif R. Mian, 2009; Jeremy C. Stein, 2002). The action  $y$  is the residual underwriting stance—for example, how soft evidence is translated into pricing, collateral, monitoring, or other terms—after verifiable rules have been imposed, and  $L$  affects only the stakes of that decision. The benchmark assumes that borrower ethnicity  $g$  is independent of  $L$  and  $s$ , and that residual uncertainty and conditional signal quality do not vary with  $L$ , producing a single size-based approval cutoff. HQ maximizes bank value, so managers of any ethnicity may favor their own group. The main-text comparison uses the bank-coethnic type  $B$  and city-coethnic type  $C$ , but the same expressions apply to any candidate  $i$ , including a third-group manager  $N$ . For conflict comparative statics,  $(\pi_i, F, \kappa)$  are fixed; for composition comparative statics, the conditional information and preference primitives are fixed. Information acquisition is match-dependent. Shared language, networks, and cultural codes make it easier to elicit and interpret a coethnic borrower’s information (Raymond Fisman, Daniel Paravisini and Vikrant Vig, 2017), while an out-group interaction can still generate recordable evidence. Only an own-group match generates the additional tacit assessment used in the communication problem, and the same match can generate favoritism (Raymond Fisman, Arkodipta Sarkar, Janis Skrastins and Vikrant Vig, 2020). Once elicited, recordable information enters the loan file and survives either authority regime; tacit information is privately observed by the manager. Conflict  $\gamma$  is a reduced-form salience index, empirically proxied by network exposure rather than necessarily local casualties. It can strengthen identity and disrupt trust, disclosure, and the interpretation of evidence across ethnic boundaries. The key asymmetry is therefore generated by a common match-level technology: cross-group recordable information deteriorates more rapidly than in-group recordable information, and the bank-coethnic manager faces the cross-group technology on a larger borrower share.

HQ appoints a manager and commits before loan information is realized to a coarse authority schedule based on manager type and verifiable loan size, but not borrower ethnicity or realized information. This restriction is load-bearing: if authority could depend on ethnicity, HQ could review only own-group loans and  $\pi_i$  would drop out of the cutoff. Under review, HQ observes the recordable assessment on every loan, receives a report about the tacit assessment on an own-group loan, chooses  $y$ , and incurs the fixed cost  $\kappa$ ; because the benefit of review scales with  $L$  while its cost does not, authority takes a loan-size cutoff form. Formal approval rights are contractible, but  $y$  and the report are not, so HQ cannot commit to a report-contingent action. We use pure-strategy perfect Bayesian equilibria in the monotone-partition class of Vincent P. Crawford and Joel Sobel (1982); Section A.4 proves that equilibrium selection does not affect the authority rule.

### A.2 A feasible match-dependent information structure

The benchmark takes the conditional residual-state distribution and signal experiments, given the manager–borrower match, to be invariant across loan sizes. For feasibility, let  $V_x, V_t \geq 0$  and  $V_\varepsilon > 0$ , with

$V = V_x + V_t + V_\varepsilon$ , and suppose

$$0 \leq \chi^O(\gamma) \leq \chi^I(\gamma) \leq V_x, \quad 0 \leq q(\gamma) \leq \sqrt{3V_t} \quad (\text{a1})$$

for every relevant conflict level.

Let the state decompose as

$$s = r + \tau + \varepsilon, \quad (\text{a2})$$

where  $r$ ,  $\tau$ , and  $\varepsilon$  are mutually independent, have mean zero, and satisfy

$$\text{Var}(r) = V_x, \quad \text{Var}(\tau) = V_t, \quad \text{Var}(\varepsilon) = V_\varepsilon > 0. \quad (\text{a3})$$

For concreteness, let  $r \sim U[-\sqrt{3V_x}, \sqrt{3V_x}]$  and  $\tau \sim U[-\sqrt{3V_t}, \sqrt{3V_t}]$ , with the natural degenerate interpretation when a component has zero variance. The component  $r$  represents information that can be elicited and recorded in a loan file;  $\tau$  represents tacit information whose posterior assessment must be communicated.

For match status  $m \in \{I, O\}$ , where  $I$  denotes an own-group borrower and  $O$  an out-group borrower, the recordable experiment produces

$$x_i^m = \mathbb{E}[r \mid \mathcal{R}_i^m], \quad x_i^m \sim U[-p^m, p^m], \quad p^m \equiv \sqrt{3\chi^m}. \quad (\text{a4})$$

The assessment  $x_i^m$  is verifiable once elicited and observed by HQ. On an own-group loan, a second experiment produces

$$t_i = \mathbb{E}[\tau \mid \mathcal{T}_i], \quad t_i \sim U[-q, q], \quad (\text{a5})$$

which is privately observed by the manager. An out-group loan generates no incremental tacit assessment.

**Lemma 1** (Bayesian feasibility). *For every  $(\chi^I, \chi^O, q)$  satisfying (a1), signal experiments satisfying (a4)–(a5) exist. Their posterior means satisfy*

$$\mathbb{E}[s \mid x_i^I, t_i, g = i] = x_i^I + t_i, \quad \mathbb{E}[s \mid x_i^O, g \neq i] = x_i^O. \quad (\text{a6})$$

*The state remains non-invertible because  $V_\varepsilon > 0$ .*

*Proof.* Let  $Z \sim U[-1, 1]$ . For  $0 \leq a \leq A$ , write  $a = \lambda A$  with  $\lambda \in [0, 1]$ . For every convex function  $\varphi$ ,

$$\varphi(aZ) = \varphi(\lambda AZ + (1 - \lambda)0) \leq \lambda \varphi(AZ) + (1 - \lambda)\varphi(0).$$

Taking expectations and applying Jensen's inequality gives  $\mathbb{E}[\varphi(aZ)] \leq \mathbb{E}[\varphi(AZ)]$ . Thus  $U[-a, a]$  is dominated by  $U[-A, A]$  in convex order. By the martingale-coupling theorem of [Volker Strassen \(1965\)](#), there is a signal experiment whose posterior mean has distribution  $U[-a, a]$ . Apply this result to  $r$  with  $(a, A) = (p^m, \sqrt{3V_x})$  for each  $m \in \{I, O\}$  and to  $\tau$  with  $(a, A) = (q, \sqrt{3V_t})$ . Independence of the components and experiments yields (a6);  $\varepsilon$  retains positive variance.  $\square$

Calibration and orthogonality imply residual loss

$$\mathbb{E}[(s - x_i^I - t_i)^2 \mid g = i] = V - \chi^I - \frac{q^2}{3}, \quad (\text{a7})$$

$$\mathbb{E}[(s - x_i^O)^2 \mid g \neq i] = V - \chi^O. \quad (\text{a8})$$

Define portfolio-average recordable information

$$\bar{\chi}_i = (1 - \pi_i)\chi^O + \pi_i\chi^I = \chi^O + \pi_i\Delta_\chi. \quad (\text{a9})$$

Averaging the residual losses gives

$$I_i = V - \bar{\chi}_i - \pi_i \frac{q^2}{3}, \quad (\text{a10})$$

which proves (3). The city-coethnic manager's recordable-information advantage follows without type-specific signal technologies:

$$\bar{\chi}_C - \bar{\chi}_B = (\pi_C - \pi_B)\Delta_\chi \geq 0. \quad (\text{a11})$$

Total information is likewise better under  $C$  whenever  $\pi_C > \pi_B$  and  $\Delta_\chi + q^2/3 > 0$ .

### A.3 Delegated decisions

On an own-group loan, manager  $i$  observes  $(x_i^I, t_i)$  and minimizes

$$\mathbb{E}[(y - s - b)^2 \mid x_i^I, t_i, g = i].$$

By (a6), her delegated action is

$$y_i^{\text{del}} = x_i^I + t_i + b. \quad (\text{a12})$$

HQ's expected loss is  $V - \chi^I - q^2/3 + b^2$ . On an out-group loan, the manager is aligned and chooses  $y = x_i^O$ , generating loss  $V - \chi^O$ . Averaging across borrower groups yields

$$\begin{aligned} \ell_i^{\text{del}}(L) &= \left[ (1 - \pi_i)(V - \chi^O) + \pi_i \left( V - \chi^I - \frac{q^2}{3} + b^2 \right) \right] L \\ &= (I_i + \pi_i b^2) L, \end{aligned} \quad (\text{a13})$$

which proves (5).

### A.4 Biased communication of tacit in-group information

Consider an own-group borrower under manager  $i$ . The recordable assessment  $x_i^I$  is observed by HQ. Conditional on  $x_i^I$ , HQ's ideal residual action is  $t_i$ , while the manager's ideal residual action is  $t_i + b$ . The communication problem therefore depends only on the privately observed tacit component  $t_i \sim U[-q, q]$ . If  $q = 0$ , this component is degenerate, no informative communication is possible, and the review loss below follows directly. For the partition derivation, take  $q > 0$ .

We characterize the standard pure-strategy monotone partition equilibria. Let

$$-q = z_0 < z_1 < \cdots < z_K = q$$

be the boundaries of a  $K$ -bin equilibrium and let  $d_k = z_k - z_{k-1}$ . Since  $t_i$  is uniform, HQ chooses the midpoint

$$m_k = \frac{z_{k-1} + z_k}{2}$$

following a report corresponding to bin  $k$ . At an interior boundary  $z_k$ , sender indifference requires

$$[m_k - (z_k + b)]^2 = [m_{k+1} - (z_k + b)]^2,$$

which is equivalent to

$$d_{k+1} = d_k + 4b. \quad (\text{a14})$$

Using  $\sum_{k=1}^K d_k = 2q$  gives

$$d_k = \frac{2q}{K} + 2b(2k - K - 1). \quad (\text{a15})$$

The smallest bin is  $d_1$ , so a genuine  $K$ -bin equilibrium exists if and only if

$$bK(K - 1) < q. \quad (\text{a16})$$

The one-bin uninformative equilibrium always exists.

Conditional on a bin of length  $d_k$ , HQ's mean-squared error in predicting  $t_i$  is  $d_k^2/12$ . The ex ante communication loss is

$$\begin{aligned} C_K(b, q) &= \sum_{k=1}^K \frac{d_k}{2q} \frac{d_k^2}{12} \\ &= \frac{q^2}{3K^2} + \frac{b^2}{3}(K^2 - 1). \end{aligned} \quad (\text{a17})$$

For every informative monotone partition equilibrium,  $K \geq 2$ . By (a16),

$$\begin{aligned} \frac{C_K}{b^2} &> \frac{(K - 1)^2 + K^2 - 1}{3} \\ &= \frac{2K(K - 1)}{3} > 1. \end{aligned} \quad (\text{a18})$$

Thus every informative equilibrium in this class creates a larger decision distortion than delegation, even before the review cost is paid, and is never chosen by HQ.

For  $K = 1$ , HQ learns nothing about  $t_i$ , so

$$C_1 = \text{Var}(t_i) = \frac{q^2}{3}.$$

If uninformative review is preferred to delegation for any loan, then necessarily

$$b^2 > \frac{q^2}{3},$$

which implies  $b > q/\sqrt{3} > q/2$ . A two-bin equilibrium requires  $2b < q$ , and equilibria with more bins require still smaller bias. Hence, whenever review can be optimal, no informative monotone partition equilibrium exists.

Under uninformative review, HQ retains  $x_i^I$  but not  $t_i$ . Expected loss on an own-group loan is therefore

$$V - \chi^I, \quad (\text{a19})$$

which exceeds the full-information residual by  $q^2/3$ . On an out-group loan,  $x_i^O$  is recordable and no strategic report is required, so loss is  $V - \chi^O$ . Averaging over borrower groups and adding  $\kappa$  gives

$$\begin{aligned}\ell_i^{\text{hq}}(L) &= [(1 - \pi_i)(V - \chi^O) + \pi_i(V - \chi^I)] L + \kappa \\ &= \left(I_i + \pi_i \frac{q^2}{3}\right) L + \kappa,\end{aligned}\tag{a20}$$

which proves (6).

**Lemma 2** (Selection robustness). *Within pure-strategy monotone-partition perfect Bayesian equilibria, the optimal authority rule is invariant to equilibrium selection. If an informative partition equilibrium exists, delegation strictly dominates review for HQ; if review is optimal, no informative partition equilibrium exists and the relevant review outcome is uninformative.*

*Proof.* For any  $K \geq 2$ , (a18) gives  $C_K > b^2$  before the review cost is paid. If uninformative review is weakly preferred for some loan, then  $b^2 > q^2/3$ , which implies  $2b > q$  and violates (a16) for every  $K \geq 2$ .  $\square$

## A.5 The authority rule and comparative statics

Subtracting (6) from (5) shows that delegation is weakly preferred precisely when

$$\pi_i \left(b^2 - \frac{q^2}{3}\right) L \leq \kappa.\tag{a21}$$

This proves the authority rule and (8). To nest the full-delegation corner, define

$$c_i(\gamma) = \begin{cases} +\infty, & h_i(\gamma) \leq 0, \\ \hat{L}_i^*(\gamma), & h_i(\gamma) > 0. \end{cases}\tag{a22}$$

Then  $D_i(L, \gamma) = \mathbf{1}\{L \leq c_i(\gamma)\}$ . Portfolio-average recordable information  $\bar{\chi}_i$  cancels because it remains available under both authority regimes.

The authority conclusions require only that the net control wedge per own-group match increases,  $d'(\gamma) > 0$ . The primitive conditions in (4) are sufficient because

$$d'(\gamma) = 2bb' - \frac{2}{3}qq' > 0.\tag{a23}$$

For any manager with  $\pi_i > 0$ ,

$$h'_i(\gamma) = \pi_i d'(\gamma) > 0.\tag{a24}$$

Therefore  $D_i(L, \gamma)$  is weakly decreasing in conflict for every  $L$ , including changes that enter or leave a corner. Whenever  $h_i > 0$ ,

$$\frac{d\hat{L}_i^*}{d\gamma} = -\frac{\hat{L}_i^*}{h_i} h'_i = -\hat{L}_i^* \frac{d'}{d} < 0.\tag{a25}$$

Extend  $F$  by zero below  $\underline{L}$ , one above  $\bar{L}$ , and set  $F(+\infty) = 1$ . Then  $A_i(\gamma) = F(c_i(\gamma))$ . For  $\gamma_2 > \gamma_1$ ,

$$A_i(\gamma_2) - A_i(\gamma_1) = F(c_i(\gamma_2)) - F(c_i(\gamma_1)) \leq 0. \quad (\text{a26})$$

The inequality is strict if and only if

$$\mathbb{P}(c_i(\gamma_2) < L \leq c_i(\gamma_1)) > 0. \quad (\text{a27})$$

Continuity of  $F$  is not needed for this monotonicity or strictness statement.

For the composition comparative static, hold  $b$  and  $q$  fixed and hence hold  $d$  fixed. If  $d \leq 0$ , then  $h_i = \pi_i d \leq 0$  for every  $\pi_i$ , so every loan is delegated and authority is constant in  $\pi_i$ . If  $d > 0$ ,  $h_i$  increases linearly in  $\pi_i$  and, for  $\pi_i > 0$ ,

$$\frac{\partial \hat{L}_i^*}{\partial \pi_i} = -\frac{\hat{L}_i^*}{\pi_i} < 0, \quad \frac{\partial \hat{L}_i^*}{\partial \kappa} = \frac{\hat{L}_i^*}{\kappa} > 0. \quad (\text{a28})$$

Thus, whenever  $d > 0$ , the cutoff  $c_i = \hat{L}_i^*$  strictly decreases in  $\pi_i$  for every  $\pi_i > 0$ . The pointwise delegation rule is weakly decreasing, and aggregate autonomy decreases strictly only if a positive mass of loans crosses the cutoff.

## A.6 Appointment and the cross-group information restriction

For the bank–city comparison, suppose the groups differ, so  $\pi_C > \pi_B$ , and define  $\Phi(\gamma) \equiv \Lambda_B(\gamma) - \Lambda_C(\gamma)$ . For each loan size, HQ selects the smaller of the additional delegated loss  $\pi_i b^2 L$  and the reviewed loss  $\pi_i q^2 L/3 + \kappa$ , on top of the residual information loss  $I_i L$ . Hence

$$\Lambda_i(\gamma) = I_i(\gamma)\mu + \mathbb{E} \left[ \min \left\{ \pi_i b(\gamma)^2 L, \pi_i \frac{q(\gamma)^2}{3} L + \kappa \right\} \right]. \quad (\text{a29})$$

Factoring out the tacit-information term gives (9). The exact appointment decomposition is

$$\begin{aligned} \Phi(\gamma) &= (\bar{\chi}_C - \bar{\chi}_B)\mu + \mathbb{E} [\min\{h_B L, \kappa\} - \min\{h_C L, \kappa\}] \\ &= (\pi_C - \pi_B)\Delta_\chi \mu + \mathbb{E} [\min\{h_B L, \kappa\} - \min\{h_C L, \kappa\}]. \end{aligned} \quad (\text{a30})$$

The first term is  $C$ 's recordable-information advantage: she uses the more resilient in-group technology on a larger borrower share. The second compares the managers' minimized control costs after HQ optimizes review.

Because  $F$  is continuous, the equality event  $h_i L = \kappa$  has probability zero. Wherever the primitives are differentiable, the envelope theorem gives

$$\Lambda'_i = -\bar{\chi}'_i \mu + h'_i M_{D,i}. \quad (\text{a31})$$

Since  $\bar{\chi}'_i = \chi'^{O'} + \pi_i \Delta'_\chi$ , the common deterioration in recordable information cancels from the appointment comparison, leaving

$$\Phi' = \mu(\pi_C - \pi_B)\Delta'_\chi + h'_B M_{D,B} - h'_C M_{D,C}. \quad (\text{a32})$$

Thus conflict shifts appointment toward  $C$  if and only if

$$\mu(\pi_C - \pi_B)\Delta'_\chi > h'_C M_{D,C} - h'_B M_{D,B}. \quad (\text{a33})$$

The left side is  $C$ 's relative information advantage as cross-group borrower communication deteriorates: because  $\pi_C > \pi_B$ , she relies on the less resilient out-group technology for a smaller portfolio share. The right side is the relative growth of her optimized control cost. Here cross-group communication means borrower–manager information acquisition, not manager–HQ cheap talk. Unlike a direct assumption on  $\chi'_B$  and  $\chi'_C$ , the asymmetry is generated by one common match-level technology and the observed difference  $\pi_C > \pi_B$ .

**A primitive dominance condition:** Under the maintained common own-group bias and tacit-information technologies,  $d = b^2 - q^2/3$ ,  $h'_i = \pi_i d'$ , and

$$\Phi' = \mu(\pi_C - \pi_B)\Delta'_\chi - d' [\pi_C M_{D,C} - \pi_B M_{D,B}]. \quad (\text{a34})$$

For every loan size,

$$\pi_C L D_C - \pi_B L D_B \leq (\pi_C - \pi_B) L. \quad (\text{a35})$$

If  $d \leq 0$ , both types delegate and the inequality holds with equality. If  $d > 0$ ,  $D_C \leq D_B$ : the only additional case has  $D_B = 1$  and  $D_C = 0$ , making the left side negative. Taking expectations in (a35) yields

$$\Phi' \geq \mu(\pi_C - \pi_B)(\Delta'_\chi - d'). \quad (\text{a36})$$

Consequently,

$$\Delta'_\chi > d' \quad (\text{a37})$$

is sufficient for conflict to favor  $C$  under any optimal authority allocation. If all loans are delegated, (a36) holds with equality, so (a37) is also necessary. The comparison poses the model's empirical appointment question: does conflict widen the in-group recordable-information premium faster than it raises the net own-group control wedge? When  $q' = 0$ , it reduces to  $\Delta'_\chi > 2bb'$ ; allowing  $q' \leq 0$  replaces  $2bb'$  with  $d' = 2bb' - 2qq'/3$  and leaves the authority result unchanged.

**Corollary (multi-type appointment ordering):** Let  $j$  and  $k$  be any candidate manager ethnicities with  $\pi_j > \pi_k$ . Repeating the pairwise argument above gives

$$\begin{aligned} \Lambda'_j - \Lambda'_k &= -\mu(\pi_j - \pi_k)\Delta'_\chi + d' [\pi_j M_{D,j} - \pi_k M_{D,k}] \\ &\leq -\mu(\pi_j - \pi_k)(\Delta'_\chi - d'). \end{aligned} \quad (\text{a38})$$

Hence (a37) makes every higher- $\pi$  candidate relatively more attractive as conflict rises, absent differential assignment-cost responses. In particular, any third-group manager  $N$  with  $\pi_N > \pi_B$  becomes more attractive relative to  $B$ ; if  $\pi_C > \pi_N > \pi_B$ , the operating-loss responses satisfy  $\Lambda'_C < \Lambda'_N < \Lambda'_B$ . The result does not sign the aggregate share of  $N$ , because  $N$  can gain appointments from  $B$  while losing them to candidates with still larger  $\pi_i$ .

## A.7 Managerial assignment costs

Let  $w_i(\gamma)$  be a differentiable reduced-form cost of assigning manager  $i$ , measured in the same branch-level units as  $\Lambda_i$  and encompassing relocation, retention, or safety premia. Define  $\Psi(\gamma) \equiv \Phi(\gamma) + w_B(\gamma) - w_C(\gamma)$ . HQ appoints  $C$  if and only if  $\Psi \geq 0$ , and conflict shifts appointment toward  $C$  if and only if

$$\mu(\pi_C - \pi_B)\Delta'_\chi + (w'_B - w'_C) > h'_C M_{D,C} - h'_B M_{D,B}. \quad (\text{a39})$$

Thus a rise in the relative relocation, retention, or safety cost of assigning  $B$  makes reallocation toward  $C$  more likely. A conflict-induced customer preference for a city-coethnic manager enters identically as a reduction in  $w_C$ . Appointment alone identifies only the composite condition (a39); it does not disentangle information dominance from assignment-cost responses. Auxiliary tests based on reassigned insiders, authority changes holding manager type fixed, and managers' local origins would bear more directly on the cross-group information interpretation.

Conditional on the appointed manager,  $w_i$  cancels from the authority comparison  $h_i L \leq \kappa$ , leaving Parts 2 and 3 of Proposition 1 unaffected. Assignment costs can nevertheless change observed branch-level authority by changing the appointed type. Holding manager type fixed removes this selection channel and provides the cleanest empirical analogue of the direct information–incentive response.

## A.8 Appointment and restricted authority can coexist

The recordable information premium allows HQ to appoint an informed local manager while reviewing some of her decisions. Review preserves the recordable assessment even when strategic communication eliminates the value of the tacit component.

**Lemma 3** (Nonempty coexistence region). *A sufficient condition for  $C$  to be preferred to  $B$  is*

$$\left[ \bar{\chi}_C - \bar{\chi}_B - \pi_B \frac{q^2}{3} \right] \mu > \kappa. \quad (\text{a40})$$

*Under the common match-level technology, the bracket equals  $(\pi_C - \pi_B)\Delta_\chi - \pi_B q^2/3$ . A sufficient condition for HQ to review a positive mass of decisions under  $C$  is*

$$h_C > 0 \quad \text{and} \quad \mathbb{P}\left(L > \frac{\kappa}{h_C}\right) > 0. \quad (\text{a41})$$

*These conditions hold jointly for a nonempty parameter region.*

*Proof.* The minimized agency and review term in  $\Lambda_B$  is nonnegative, so

$$\Lambda_B \geq I_B \mu = \left( V - \bar{\chi}_B - \pi_B \frac{q^2}{3} \right) \mu.$$

Under  $C$ , reviewing every loan is feasible. Hence

$$\Lambda_C \leq (V - \bar{\chi}_C) \mu + \kappa.$$

Condition (a40) therefore implies  $\Lambda_C < \Lambda_B$ . Condition (a41) places positive loan mass above  $C$ 's approval cutoff. Joint feasibility follows by taking the recordable match premium  $\Delta_\chi$  sufficiently large,  $\kappa$  sufficiently small, and  $b$  sufficiently large. HQ can then appoint  $C$  for recordable information obtained from local borrowers while limiting the problems created by her tacit information and bias through review.  $\square$

A positive recordable match premium separates the informational value of appointment from the informational cost of review. If  $\Delta_\chi = 0$  and all of  $C$ 's relative information advantage is tacit, reviewing own-group loans destroys that advantage and the earlier appointment–centralization incompatibility can reappear.

## **B Inference and bank-time comparisons**

### **B.1 Manager ethnicity and delegation with bank-time fixed effects**

Table [B1](#) compares branches in bank-coethnic and non-bank-coethnic cities within the same bank and period. For this exercise, Network Exposure is deliberately measured over the bank’s complete branch network, including conflict near the focal branch. It therefore varies only at the bank-time level and its uninteracted main effect is absorbed by bank-time fixed effects. Panel A reports coefficients on the interaction between this exposure and an indicator for a non-bank-coethnic city. The four outcomes are the three manager types and delegation. All specifications include branch and bank-time fixed effects and allow non-bank-coethnic branches to follow a separate time trend. The city-coethnic manager estimate is 0.128 (s.e. 0.030), while the delegation estimate is  $-2,543$  thousand 2018 ETB (s.e. 934). Panel B relates delegation to manager ethnicity and the local population share belonging to the manager’s ethnic group separately in bank-city non-coethnic and coethnic locations.

Table B1: Manager ethnicity and delegation with bank-time fixed effects

| Panel A: Manager ethnicity and delegation                    |                              |                              |                                 |                        |                           |                         |
|--------------------------------------------------------------|------------------------------|------------------------------|---------------------------------|------------------------|---------------------------|-------------------------|
|                                                              | (1)<br>City-coethnic manager | (2)<br>Bank-coethnic manager | (3)<br>Neither-coethnic manager | (4)<br>Delegation      |                           |                         |
| Network Exposure ×<br>Non-coethnic city                      | 0.128***<br>(0.030)          | -0.074<br>(0.060)            | -0.012<br>(0.036)               | -2543**<br>(934)       |                           |                         |
| Branch FE                                                    | Yes                          | Yes                          | Yes                             | Yes                    |                           |                         |
| Bank-time FE                                                 | Yes                          | Yes                          | Yes                             | Yes                    |                           |                         |
| Non-bank-coethnic-<br>time FE                                | Yes                          | Yes                          | Yes                             | Yes                    |                           |                         |
| 2018 Mean dep. var                                           | 0.411                        | 0.729                        | 0.171                           | 1755                   |                           |                         |
| N (Branches)                                                 | 978                          | 978                          | 978                             | 553                    |                           |                         |
| Banks                                                        | 16                           | 16                           | 16                              | 16                     |                           |                         |
| Panel B: Delegation, manager type and local population share |                              |                              |                                 |                        |                           |                         |
|                                                              | Bank and city not coethnic   |                              |                                 | Bank and city coethnic |                           |                         |
|                                                              | (1)<br>Manager type          | (2)<br>+ population share    | (3)<br>Population share         | (4)<br>Manager type    | (5)<br>+ population share | (6)<br>Population share |
| City-coethnic<br>manager                                     | -473.3<br>(1012.2)           | 1211.0<br>(1544.9)           |                                 | -2423.9**<br>(1041.8)  | -767.6<br>(1494.4)        |                         |
| Neither-coethnic<br>manager                                  | 736.6<br>(1404.2)            | 944.2<br>(1502.2)            |                                 |                        |                           |                         |
| Manager-group<br>population share                            |                              | -4147.1**<br>(1894.3)        | -3200.4***<br>(1051.9)          |                        | -3182.1<br>(1901.5)       | -4067.5**<br>(1480.1)   |
| Branch FE                                                    | Yes                          | Yes                          | Yes                             | Yes                    | Yes                       | Yes                     |
| Bank-time FE                                                 | Yes                          | Yes                          | Yes                             | Yes                    | Yes                       | Yes                     |
| 2018 Mean dep.<br>var                                        | 1688.5                       | 1688.5                       | 1688.5                          | 1849.3                 | 1849.3                    | 1849.3                  |
| N (Branches)                                                 | 321                          | 321                          | 321                             | 229                    | 229                       | 229                     |
| Banks                                                        | 15                           | 15                           | 15                              | 13                     | 13                        | 13                      |

*Notes:* Panel A uses the whole-network measure of conflict exposure. This measure includes conflict near the focal branch and is constant within bank-time cells, so its main effect is absorbed. The reported coefficients are on the interaction between Network Exposure and an indicator for a branch in a city whose ethnicity differs from the bank's. The first three outcomes are indicators; delegation in column 4 is measured in thousands of 2018 ETB. Panel A includes non-bank-coethnic-time fixed effects to allow the two groups to follow different time trends. Panel B regresses delegation on manager type and the local population share belonging to the manager's ethnic group. Columns 1–3 report results for the non-bank-coethnic sample, and columns 4–6 report results for the bank-city-coethnic sample. Within each group, the columns include manager-type indicators, manager type plus the population share, and the population share alone. City-coethnic manager and Neither-coethnic manager are indicators. In columns 1–2, bank-coethnic managers are the omitted category. In columns 4–5, the bank's and city's ethnicities coincide, so managers coethnic with neither are the omitted category. Delegation is the largest loan a manager can approve without headquarters' approval, measured in thousands of 2018 ETB and winsorized at the 95th percentile within cells defined by time and bank-city coethnicity. All specifications include branch and bank-time fixed effects. Standard errors are clustered by bank and reported in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

## B.2 Delegation results for branches with no change in manager

Table B2: Delegation among branches with an unchanged manager type

|                                            | (1)<br>All stable<br>types | (2)<br>Bank or city<br>coethnic | (3)<br>Bank or city<br>coethnic | (4)<br>Bank or city<br>coethnic |
|--------------------------------------------|----------------------------|---------------------------------|---------------------------------|---------------------------------|
| <i>Panel A: Conflict exposure</i>          |                            |                                 |                                 |                                 |
| Network Exposure                           | -811.1<br>(539.9)          | -749.7<br>(567.8)               | -328.8<br>(1088.2)              | -442.1<br>(1075.6)              |
| Network Exposure $\times$ City-coethnic    |                            |                                 | -691.6<br>(1100.3)              |                                 |
| Network Exposure $\times$ Population share |                            |                                 |                                 | -614.6<br>(1321.4)              |
| 2018 mean dep. var.                        | 1709.7                     | 1700.0                          | 1700.0                          | 1700.0                          |
| Branch-years                               | 248                        | 240                             | 240                             | 240                             |
| Branches                                   | 124                        | 120                             | 120                             | 120                             |
| <i>Panel B: 2018–2022 difference</i>       |                            |                                 |                                 |                                 |
| 2022                                       | 6565.1***<br>(838.6)       | 6781.4***<br>(859.3)            | 8868.0***<br>(1236.8)           | 10307.2***<br>(1417.5)          |
| 2022 $\times$ City-coethnic                |                            |                                 | -5007.7***<br>(1582.9)          |                                 |
| 2022 $\times$ Population share             |                            |                                 |                                 | -8628.1***<br>(1932.0)          |
| 2018 mean dep. var.                        | 1709.7                     | 1700.0                          | 1700.0                          | 1700.0                          |
| Branch-years                               | 248                        | 240                             | 240                             | 240                             |
| Branches                                   | 124                        | 120                             | 120                             | 120                             |

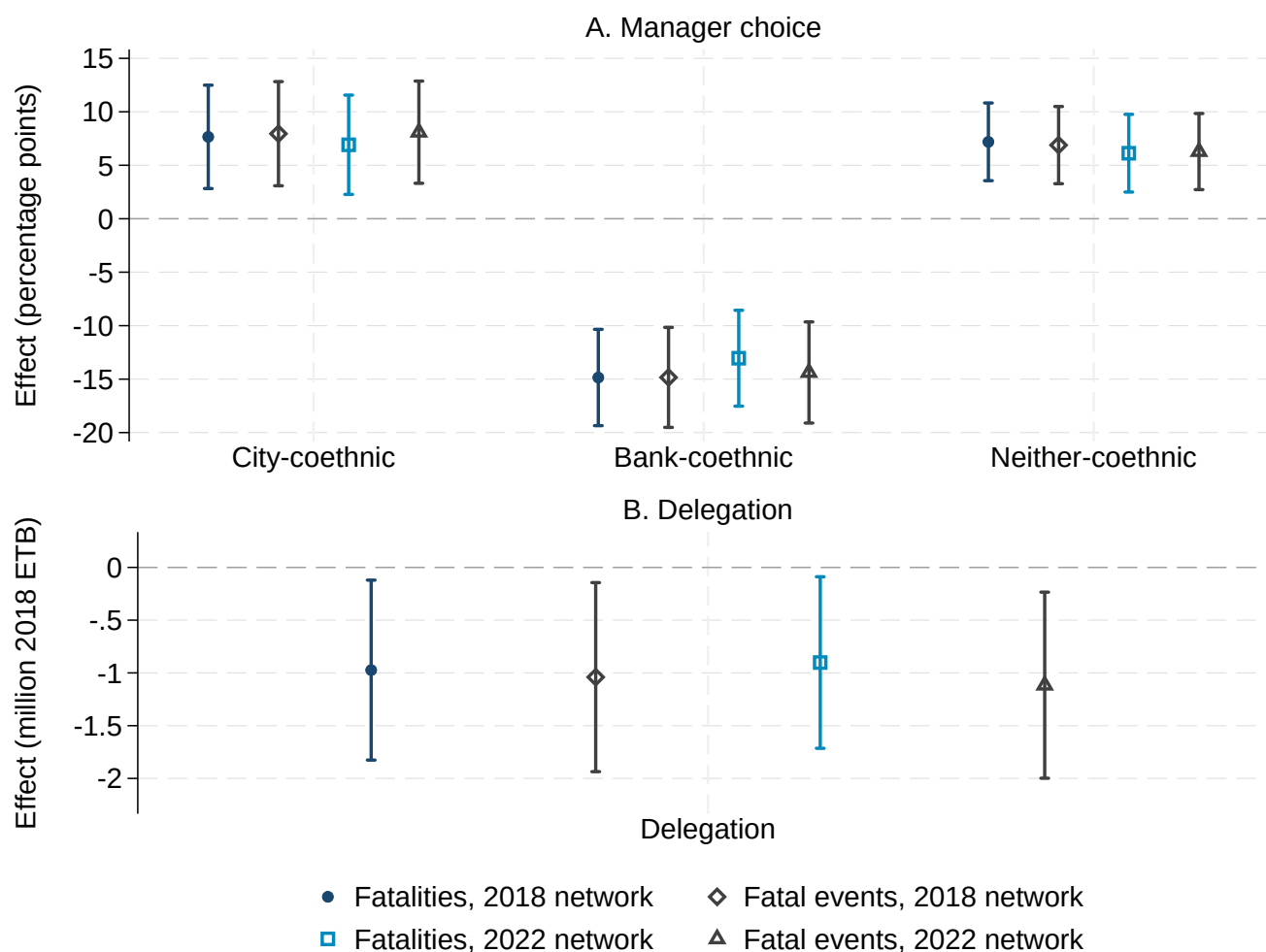
*Notes:* The sample is branch-years in cities not coethnic with the bank, restricted to branches whose manager type is unchanged between 2018 and 2022. The dependent variable in both panels is the largest loan a branch manager can approve without headquarters' approval, measured in thousands of 2018 ETB and winsorized as in main-text Table 3. *Panel A* reports the effect of network conflict exposure, the log intensity of branch-weighted exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest. *Panel B* reports the difference between the two waves and uses no conflict measure; it asks whether the two groups of branches diverged at all, rather than whether they diverged in proportion to realized exposure. Column 1 includes all stable manager types and reproduces the estimate quoted in the main text. Columns 2–4 restrict to branches whose manager is stably coethnic with either the bank or the city, excluding the four branches whose manager is coethnic with neither. Column 3 interacts the regressor with an indicator for a city-coethnic manager. Column 4 interacts it with the 2007 census population share of the manager's own ethnic group, the empirical counterpart of  $\pi_i$ ; these are population shares rather than customer shares, since the survey records no customer ethnicity. Because manager type and the population share are time-invariant within branch in this sample, their levels are collinear with the branch fixed effects and are not estimable here; only the interactions reported are identified. All columns include branch fixed effects, and Panel A additionally includes wave fixed effects. Standard errors clustered at the branch level are reported in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

## C Robustness Tests

### C.1 Alternative conflict intensity and branch-network definitions

Figure C1 implements the two alternative definitions described in the main text for both manager choice and delegation. We replace reported fatalities with the number of distinct fatal ACLED events and, separately, replace the fixed 2018 branch network with the fixed network of branches observed in 2022. The fourth series changes both choices at once. To compare estimates across differently scaled exposures, the figure reports effects of a one-standard-deviation within-branch change in each log exposure. The appointment results are stable across all four definitions: network conflict raises the probability of a city-coethnic or neither-coethnic manager and lowers the probability of a bank-coethnic manager. Delegation also falls under all four definitions.

Figure C1: Manager choice and delegation under alternative conflict and network definitions

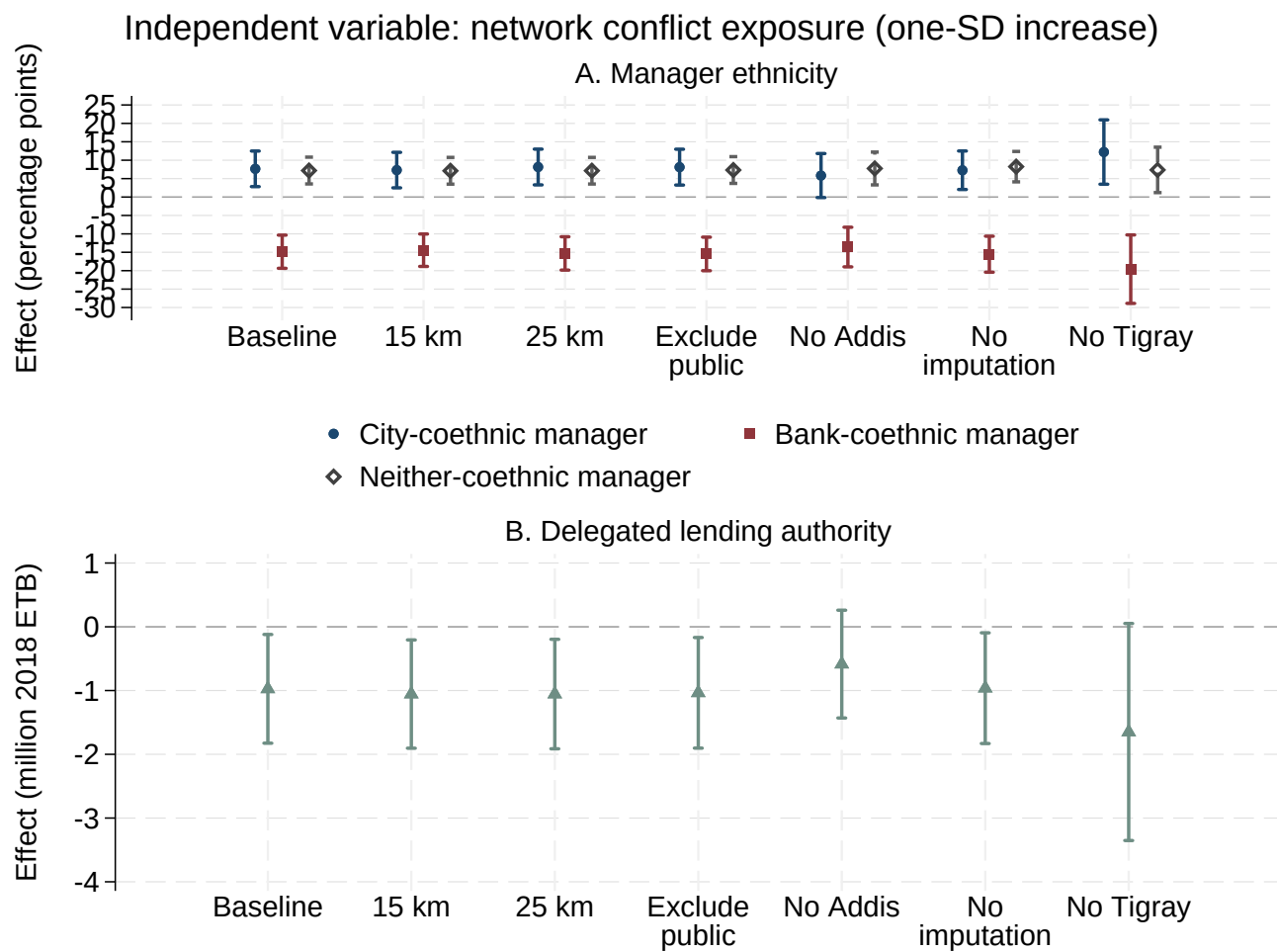


*Notes:* Panel A reports separate regressions for the three mutually exclusive manager-ethnicity outcomes among branches in non-bank-coethnic cities. Panel B reports delegation in millions of 2018 ETB. The four series form a  $2 \times 2$  comparison of conflict intensity (reported fatalities or the number of distinct ACLED events with at least one reported fatality) and network vintage (the fixed 2018 or fixed 2022 survey branch network). The network for each vintage is held fixed across both conflict periods; the 2022 variant therefore changes the network footprint and branch weights, not the network contemporaneously with conflict. All definitions use the same actor-ethnicity mapping, 20 km event-to-city match, branch-count weights, 2015–2018 and 2019–2022 windows, denominator equal to the number of bank branches outside the focal city, and strict exclusion of every event within 20 km of the focal city. Exposure is transformed as  $\ln(x + 0.01)$ ; no inverse-hyperbolic-sine transformation is used. Each regression includes branch and wave fixed effects. Standard errors are clustered by branch. Markers and 95% confidence intervals are scaled by the definition-specific standard deviation of the within-branch exposure change, calculated in the common manager-choice sample. Panel A contains 1,062 branch-wave observations from 531 branches; Panel B contains 644 observations from 322 balanced branches. The explicit fixed-2018 construction reproduces the stored headline exposure exactly.

## C.2 Manager-ethnicity and delegated-lending-authority robustness

Figure C2 collects the robustness of the manager-appointment and delegation results to the sample and measurement choices described in the main text. Moving across the horizontal axis, we vary the set  $\mathcal{E}_{bt}$  in equation 13 by matching conflict within 15 and then 25 kilometers of a branch of bank  $b$  rather than 20; we exclude the state-owned banks, CBE and CBB (“Exclude public”); we exclude branches in Addis Ababa (“No Addis”); we drop branches whose manager ethnicity is imputed rather than observed (“No imputation”, see Section I); and we drop branches in Tigray (“No Tigray”). Panel A reports the three manager-ethnicity outcomes and Panel B the delegated lending authority. The estimates are stable across every variant, both quantitatively and qualitatively: none of these choices drives the results. Table C1 reports the remaining check, dropping the branches of banks of each ethnic group one at a time, which does not fit the common scale of the figure.

Figure C2: Manager ethnicity and delegated lending authority: selected robustness checks



*Notes:* The independent variable is log bank-network conflict exposure involving the bank’s ethnic group and excluding conflict near the focal branch; markers give effects of a one-standard-deviation increase. Regressions include branch and period fixed effects in non-bank-coethnic cities; whiskers are 95% confidence intervals using branch-clustered standard errors. Panel A reports manager-ethnicity outcomes in percentage points (531 baseline branches). Panel B reports the maximum loan a manager may approve without headquarters’ approval, in millions of 2018 ETB (322 balanced baseline branches). “Exclude public” drops CBE and CBB; “No Tigray” drops branches, not Tigray-related events. The Addis and non-Addis delegation slopes are not statistically distinguishable with group-specific period effects ( $p = 0.80$ ).

Table C1: Robustness: dropping individual ethnic groups

| Panel A: Branches in Bank-Coethnic Cities |                       |                   |                   |                       |                  |                   |                   |                  |  |
|-------------------------------------------|-----------------------|-------------------|-------------------|-----------------------|------------------|-------------------|-------------------|------------------|--|
|                                           | City coethnic manager |                   |                   | Bank coethnic manager |                  |                   |                   |                  |  |
|                                           | Drop Oromo            | Drop Tigray       | Drop Amhara       | Drop SNNP             | Drop Oromo       | Drop Tigray       | Drop Amhara       | Drop SNNP        |  |
| Network Exposure                          | 0.010<br>(0.034)      | -0.005<br>(0.026) | -0.087<br>(0.053) | 0.023<br>(0.031)      | 0.010<br>(0.034) | -0.005<br>(0.026) | -0.087<br>(0.053) | 0.023<br>(0.031) |  |
| Branch FE                                 | Yes                   | Yes               | Yes               | Yes                   | Yes              | Yes               | Yes               | Yes              |  |
| Time FE                                   | Yes                   | Yes               | Yes               | Yes                   | Yes              | Yes               | Yes               | Yes              |  |
| 2018 Mean dep. var                        | 0.669                 | 0.679             | 0.698             | 0.681                 | 0.669            | 0.679             | 0.698             | 0.681            |  |
| N (Branches)                              | 347                   | 448               | 129               | 420                   | 347              | 448               | 129               | 420              |  |

| Panel B: Branches in Non-Bank-Coethnic Cities |                       |                     |                     |                       |                      |                      |                      |                      |  |
|-----------------------------------------------|-----------------------|---------------------|---------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|--|
|                                               | City coethnic manager |                     |                     | Bank coethnic manager |                      |                      |                      |                      |  |
|                                               | Drop Oromo            | Drop Tigray         | Drop Amhara         | Drop SNNP             | Drop Oromo           | Drop Tigray          | Drop Amhara          | Drop SNNP            |  |
| Network Exposure                              | 0.030*<br>(0.018)     | 0.079***<br>(0.029) | 0.048***<br>(0.016) | 0.055***<br>(0.018)   | -0.062***<br>(0.016) | -0.127***<br>(0.031) | -0.097***<br>(0.015) | -0.112***<br>(0.017) |  |
| Branch FE                                     | Yes                   | Yes                 | Yes                 | Yes                   | Yes                  | Yes                  | Yes                  | Yes                  |  |
| Time FE                                       | Yes                   | Yes                 | Yes                 | Yes                   | Yes                  | Yes                  | Yes                  | Yes                  |  |
| 2018 Mean dep. var                            | 0.178                 | 0.197               | 0.222               | 0.156                 | 0.775                | 0.755                | 0.735                | 0.803                |  |
| p-value                                       | 0.605                 | 0.031               | 0.014               | 0.377                 | 0.056                | 0.003                | 0.857                | 0.000                |  |
| N (Branches)                                  | 404                   | 458                 | 275                 | 456                   | 404                  | 458                  | 275                  | 456                  |  |

Notes: The dependent variable is an indicator that equals 1 when a branch's manager is coethnic with the ethnic group associated with the city (columns 1–4) or with the ethnic group associated with the branch's bank (columns 5–8). Each column drops all branches belonging to banks of the indicated ethnic group. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 13. Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. The reported  $p$ -value is for the test that the slope on Network Exposure is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

### C.3 The effect of conflict on employees being local

Table C2: Bank-level exposure to ethnic conflict and local staff

| Panel A: Branches in Bank-Coethnic Cities     |                        |                   |                   |                       |                    |                     |
|-----------------------------------------------|------------------------|-------------------|-------------------|-----------------------|--------------------|---------------------|
|                                               | Manager born in region |                   |                   | Share local employees |                    |                     |
|                                               | (1)                    | (2)               | (3)               | (4)                   | (5)                | (6)                 |
| Network Exposure                              | -0.037<br>(0.030)      | -0.107<br>(0.094) | -0.037<br>(0.030) | -0.027*<br>(0.014)    | 0.013<br>(0.051)   | -0.027**<br>(0.014) |
| Local Exposure                                |                        |                   | 0.008<br>(0.012)  |                       |                    | 0.012**<br>(0.005)  |
| Branch FE                                     | Yes                    | Yes               | Yes               | Yes                   | Yes                | Yes                 |
| Time FE                                       | Yes                    | No                | Yes               | Yes                   | No                 | Yes                 |
| City-Time FE                                  | No                     | Yes               | No                | No                    | Yes                | No                  |
| 2018 Mean dep. var                            | 0.421                  | 0.421             | 0.421             | 0.556                 | 0.556              | 0.556               |
| N (Branches)                                  | 214                    | 214               | 214               | 448                   | 448                | 448                 |
| Panel B: Branches in Non-Bank-Coethnic Cities |                        |                   |                   |                       |                    |                     |
|                                               | Manager born in region |                   |                   | Share local employees |                    |                     |
|                                               | (1)                    | (2)               | (3)               | (4)                   | (5)                | (6)                 |
| Network Exposure                              | -0.025<br>(0.027)      | -0.026<br>(0.028) | -0.024<br>(0.027) | -0.008<br>(0.010)     | -0.018*<br>(0.010) | -0.009<br>(0.010)   |
| Local Exposure                                |                        |                   | -0.006<br>(0.018) |                       |                    | 0.020***<br>(0.005) |
| Branch FE                                     | Yes                    | Yes               | Yes               | Yes                   | Yes                | Yes                 |
| Time FE                                       | Yes                    | No                | Yes               | Yes                   | No                 | Yes                 |
| City-Time FE                                  | No                     | Yes               | No                | No                    | Yes                | No                  |
| 2018 Mean dep. var                            | 0.448                  | 0.448             | 0.448             | 0.584                 | 0.584              | 0.584               |
| p-value                                       | 0.762                  | 0.413             | 0.741             | 0.265                 | 0.551              | 0.279               |
| N (Branches)                                  | 250                    | 250               | 250               | 531                   | 531                | 531                 |

*Notes:* This table shows the results of a regression on two measures of whether staff comes from the same region as the branch. Share local employees is the share of employees who were born in the same city as the branch, local manager is a dummy for whether the manager was born in the same region as the branch. Note that this question was added to the survey after around 500 branches were interviewed in 2018 resulting in a small sample size. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 13. Local exposure is the log intensity of conflict within 20 km of the branch's city, defined in Equation 14. The mean (SD) of the within-branch change in log Network Exposure from 2018 to 2022 is 0.955 (1.340). Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. The reported  $p$ -value is for the test that the slope on Network Exposure is identical in bank-coethnic and non bank-coethnic cities. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

## C.4 Alternative transformation of the conflict variables

In this section, we report the results from the specification in equation 15 using an asinh transformation ( $\text{asinh}(x) = \ln(x + \sqrt{x^2 + 1})$ ) of the network exposure variable

Table C3: IHS Robustness: Branch Managers

| Panel A: Bank-City Coethnic     |                                 |                                 |                         |                          |                            |                     |
|---------------------------------|---------------------------------|---------------------------------|-------------------------|--------------------------|----------------------------|---------------------|
|                                 | (1)<br>City coethnic<br>manager | (2)<br>Bank coethnic<br>manager | (3)<br>Delegation       | (4)<br>Tenure in<br>bank | (5)<br>Tenure in<br>branch | (6)<br>Age          |
| IHS Network Exposure            | -0.009<br>(0.027)               | -0.009<br>(0.027)               | -120.900<br>(474.728)   | 0.051<br>(0.249)         | -0.340**<br>(0.140)        | -0.033<br>(0.386)   |
| Branch FE                       | Yes                             | Yes                             | Yes                     | Yes                      | Yes                        | Yes                 |
| Time FE                         | Yes                             | Yes                             | Yes                     | Yes                      | Yes                        | Yes                 |
| 2018 Mean dep. var              | 0.680                           | 0.680                           | 1852.814                | 6.859                    | 2.390                      | 33.825              |
| N                               | 447                             | 447                             | 231                     | 447                      | 446                        | 417                 |
| Panel B: Not Bank-City Coethnic |                                 |                                 |                         |                          |                            |                     |
|                                 | City coethnic<br>manager        | Bank coethnic<br>manager        | Delegation              | Tenure in<br>bank        | Tenure in<br>branch        | Age                 |
| IHS Network Exposure            | 0.055***<br>(0.017)             | -0.105***<br>(0.016)            | -689.922**<br>(312.323) | 0.538***<br>(0.167)      | 0.231**<br>(0.103)         | 0.586***<br>(0.220) |
| Branch FE                       | Yes                             | Yes                             | Yes                     | Yes                      | Yes                        | Yes                 |
| Time FE                         | Yes                             | Yes                             | Yes                     | Yes                      | Yes                        | Yes                 |
| 2018 Mean dep. var              | 0.185                           | 0.770                           | 1684.783                | 6.837                    | 2.366                      | 33.700              |
| p-value                         | 0.050                           | 0.003                           | 0.317                   | 0.105                    | 0.001                      | 0.164               |
| N                               | 531                             | 531                             | 322                     | 528                      | 528                        | 500                 |

*Notes:* Each column reports a separate regression using the inverse hyperbolic sine transformation of Network Exposure in place of the log transformation used in the main text. The delegation outcome is the largest loan a branch manager can approve without headquarters' approval, expressed in 2018 prices and winsorized at the 95th percentile within period-by-bank-city-coethnicity cells. Panel A restricts the sample to branches in coethnic cities, while Panel B focuses on branches in non-coethnic cities. All specifications include branch and time fixed effects. Standard errors, clustered at the branch level, are reported in parentheses. The reported  $p$ -value is for the  $t$ -test that the parameter estimates for branches in bank-coethnic cities (Panel A) and in non bank-coethnic cities (Panel B) are the same. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

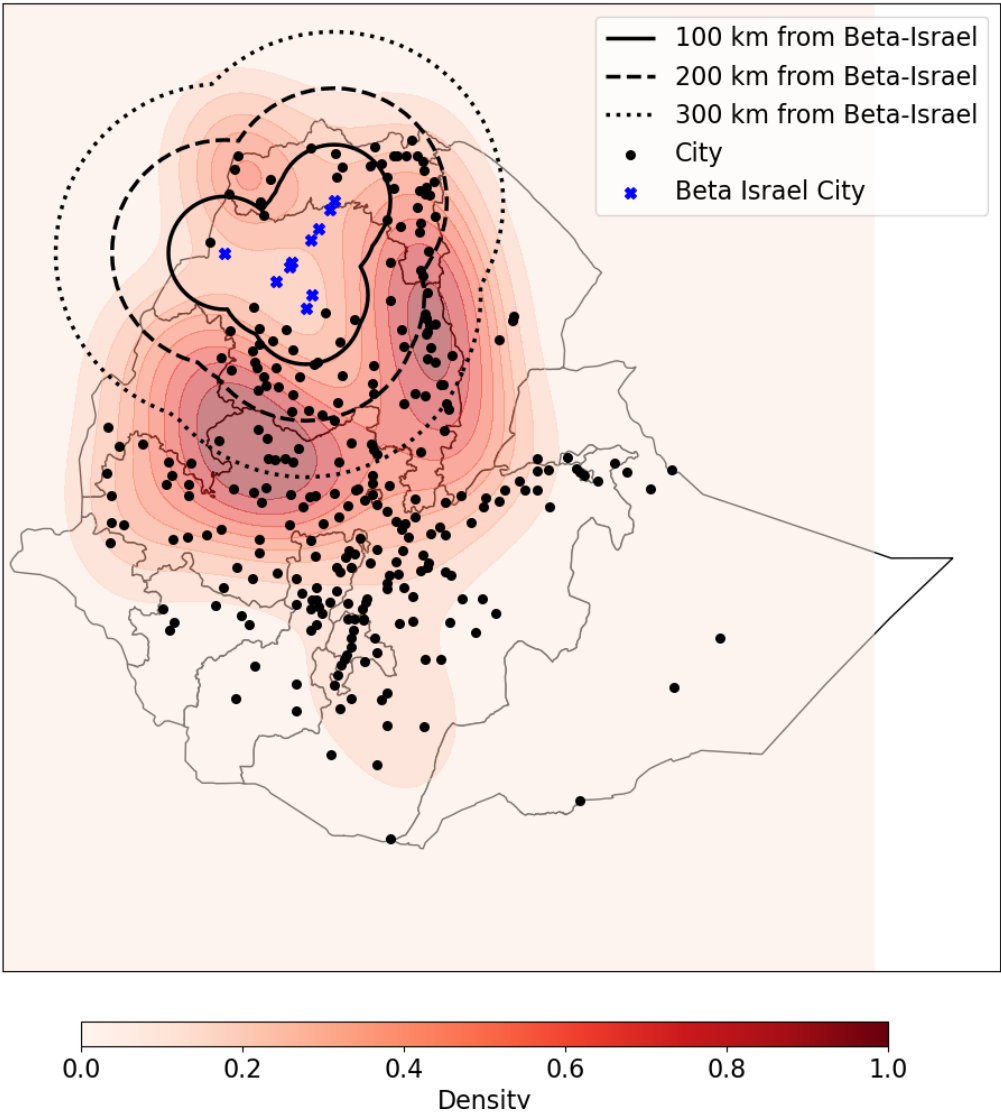
Table C4: IHS Robustness: Lending Activities

| Panel A: Bank-City Coethnic     |                       |                      |                     |                     |                   |                   |
|---------------------------------|-----------------------|----------------------|---------------------|---------------------|-------------------|-------------------|
|                                 | (1)<br>Oper. distance | (2)<br>Num. loans    | (3)<br>Loan size    | (4)<br>Lending rate | (5)<br>Collateral | (6)<br>Default    |
| IHS Network Exposure            | -2.016***<br>(0.610)  | 0.074<br>(0.120)     | 0.527***<br>(0.159) | 0.002<br>(0.001)    | 0.016<br>(0.014)  | -0.027<br>(0.020) |
| Branch FE                       | Yes                   | Yes                  | Yes                 | Yes                 | Yes               | Yes               |
| Time FE                         | Yes                   | Yes                  | Yes                 | Yes                 | Yes               | Yes               |
| 2018 Mean dep. var              | 7.818                 | 2.181                | 6.041               | 0.147               | 0.986             | 0.052             |
| N                               | 308                   | 224                  | 222                 | 292                 | 202               | 273               |
| Panel B: Not Bank-City Coethnic |                       |                      |                     |                     |                   |                   |
|                                 | Oper. distance        | Num. loans           | Loan size           | Lending rate        | Collateral        | Default           |
| IHS Network Exposure            | -0.073<br>(0.168)     | -0.248***<br>(0.056) | 0.151<br>(0.144)    | -0.000<br>(0.001)   | -0.006<br>(0.009) | -0.008<br>(0.008) |
| Branch FE                       | Yes                   | Yes                  | Yes                 | Yes                 | Yes               | Yes               |
| Time FE                         | Yes                   | Yes                  | Yes                 | Yes                 | Yes               | Yes               |
| 2018 Mean dep. var              | 7.542                 | 1.928                | 6.255               | 0.148               | 0.995             | 0.045             |
| p-value                         | 0.002                 | 0.015                | 0.080               | 0.337               | 0.196             | 0.359             |
| N                               | 413                   | 307                  | 304                 | 366                 | 267               | 325               |

*Notes:* Each column reports a separate regression using the inverse hyperbolic sine transformation of Network Exposure in place of the log transformation used in the main text. The dependent variables are the operational distance of the branch, the log of the number of loans outstanding, the log of the average loan size (deflated to 2018 prices), the lending rate on a typical loan, the percentage of collateral on a typical loan, and the share of loans that default. Panel A restricts the sample to branches in coethnic cities, while Panel B focuses on branches in non-coethnic cities. All specifications include branch and time fixed effects. Standard errors, clustered at the branch level, are reported in parentheses. The reported  $p$ -value is for the  $t$ -test that the parameter estimates for branches in bank-coethnic cities (Panel A) and in non bank-coethnic cities (Panel B) are the same. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

C.5 Beta Israel cities and conflict geography

Figure C3: Beta Israel cities and conflict contours



Notes: Blue crosses mark cities historically associated with Beta Israel communities, and black dots mark other cities. The solid, dashed, and dotted black contours indicate locations within 100, 200, and 300 km, respectively, of the nearest historical Beta Israel city. The red shading displays the density of fatalities from ethnic conflict between 2015 and 2022, with darker areas indicating greater density.

## C.6 Beta-Israel exposure and lending

Table C5 reports the lending results using the preferred 100 km Beta-Israel exposure measure. The pattern is similar to that in the baseline specification, although it highlights greater adjustment to loan terms in bank-coethnic cities. In bank-coethnic cities, exposure to Beta-Israel cities is associated with larger average loan sizes, higher collateral requirements, and lower default rates, consistent with a shift toward safer, collateral-backed lending. In non-bank-coethnic cities, banks expand their operational radius and provide fewer, larger loans. The estimates suggest that banks can continue lending effectively outside cities where the majority of customers share the bank's ethnicity.

Table C5: Bank-level exposure to Beta-Israel cities and lending (100 km)

| Panel A: Bank and city coethnic     |                       |                     |                     |                     |                     |                      |
|-------------------------------------|-----------------------|---------------------|---------------------|---------------------|---------------------|----------------------|
|                                     | (1)<br>Oper. distance | (2)<br>Num. loans   | (3)<br>Loan size    | (4)<br>Lending rate | (5)<br>Collateral   | (6)<br>Default       |
| Beta Israel Exposure                | -4.642***<br>(1.501)  | 0.040<br>(0.289)    | 1.512***<br>(0.531) | 0.000<br>(0.004)    | 0.232***<br>(0.047) | -0.128***<br>(0.048) |
| Branch FE                           | Yes                   | Yes                 | Yes                 | Yes                 | Yes                 | Yes                  |
| Time FE                             | Yes                   | Yes                 | Yes                 | Yes                 | Yes                 | Yes                  |
| 2018 Mean dep. var                  | 7.818                 | 2.181               | 6.041               | 0.147               | 0.986               | 0.052                |
| N                                   | 308                   | 224                 | 222                 | 292                 | 202                 | 273                  |
| Panel B: Bank and city not coethnic |                       |                     |                     |                     |                     |                      |
|                                     | Oper. distance        | Num. loans          | Loan size           | Lending rate        | Collateral          | Default              |
| Beta Israel Exposure                | 7.116***<br>(1.284)   | -0.768**<br>(0.364) | 3.053***<br>(0.772) | 0.002<br>(0.006)    | -0.084<br>(0.052)   | 0.056<br>(0.059)     |
| Branch FE                           | Yes                   | Yes                 | Yes                 | Yes                 | Yes                 | Yes                  |
| Time FE                             | Yes                   | Yes                 | Yes                 | Yes                 | Yes                 | Yes                  |
| 2018 Mean dep. var                  | 7.542                 | 1.928               | 6.255               | 0.148               | 0.995               | 0.045                |
| p-value                             | 0.000                 | 0.083               | 0.101               | 0.749               | 0.000               | 0.016                |
| N                                   | 413                   | 307                 | 304                 | 366                 | 267                 | 325                  |

*Notes:* The dependent variables are the operational distance of the branch, the log number of outstanding loans, the log average loan size, the lending rate on a typical loan, the percentage of collateral on a typical loan, and the share of loans in default. Beta Israel Exposure is the share of each bank's 2018 branches within 100 km of a historic Beta-Israel city, normalized to lie in  $[0, 1]$  and interacted with a post-2018 indicator. Panel A restricts the sample to branches in bank-coethnic cities; Panel B focuses on branches in non-bank-coethnic cities. Standard errors, clustered at the branch level, are reported in parentheses. The reported  $p$ -value tests whether the coefficient is identical across panels. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

## C.7 Robustness: 50 km Beta-Israel exposure window

Section 4.5 measures bank-level Beta-Israel exposure as the share of a bank's branches within 100 km of a historic Beta-Israel city. We use 100 km because the contested 1992 borders that link the Beta-Israel migration to subsequent ethnic conflict extend that far from the cities themselves. To check whether the radius drives our results, we re-estimate Table 8 and Table C5 using a tighter 50 km window. We construct  $\text{share\_branch\_beta\_50}_b$  analogously and apply the same normalization.

Table C6 reports the manager results. The patterns mirror the 100 km benchmark. In non bank-coethnic cities, exposure raises the share of city-coethnic managers and lowers the share of bank-coethnic managers; these effects, together with the reduction in delegation, all remain significant at the 1% level. The point estimates are 75–80% of the 100 km magnitudes, since the tighter radius sets many branches' exposure to zero and so removes variation. The  $p$ -value for test of a different effect in bank coethnic cities for the city-coethnic outcome rises from 0.029 to 0.052, while the bank-coethnic interaction stays highly significant ( $p = 0.003$ ).

Table C7 reports the lending results. Bank-coethnic and non bank-coethnic markets again diverge: in coethnic cities, exposure shifts lending toward larger, more collateralized, lower-default loans; in non-coethnic cities, branches serve customers over a wider area and grant larger loans. Every coefficient retains its sign, and all but two retain their significance level. We keep 100 km as the preferred specification because it more closely matches the geographic span of the contested 1992 borders.

Table C8 tightens the window further, reporting the in-city definition – the share of a bank's 2018 branches located in a historic Beta Israel city – alongside the 50 km definition for the same set of outcomes. No other sampled city lies within 20 km of a Beta Israel city, so the in-city measure is also the 20 km measure in this sample. The manager-appointment and delegation results survive at the tightest definition available.

Table C6: Bank-level exposure to Beta-Israel cities and branch managers (50 km)

| Panel A: Bank-City Coethnic     |                                 |                                 |                                 |                    |                          |                            |
|---------------------------------|---------------------------------|---------------------------------|---------------------------------|--------------------|--------------------------|----------------------------|
|                                 | (1)<br>City coethnic<br>manager | (2)<br>Bank coethnic<br>manager | (3)<br>Coethnic with<br>neither | (4)<br>Delegation  | (5)<br>Tenure in<br>bank | (6)<br>Tenure in<br>branch |
| Beta Israel Exposure            | 0.020<br>(0.090)                | 0.020<br>(0.090)                | -0.020<br>(0.090)               | -1498<br>(1944)    | -1.781*<br>(0.998)       | -1.319***<br>(0.448)       |
| Branch FE                       | Yes                             | Yes                             | Yes                             | Yes                | Yes                      | Yes                        |
| Time FE                         | Yes                             | Yes                             | Yes                             | Yes                | Yes                      | Yes                        |
| 2018 Mean dep. var              | 0.680                           | 0.680                           | 0.320                           | 1852.8             | 6.859                    | 2.390                      |
| N                               | 447                             | 447                             | 447                             | 231                | 447                      | 446                        |
| Panel B: Not Bank-City Coethnic |                                 |                                 |                                 |                    |                          |                            |
|                                 | City coethnic<br>manager        | Bank coethnic<br>manager        | Coethnic with<br>neither        | Delegation         | Tenure in<br>bank        | Tenure in<br>branch        |
| Beta Israel Exposure            | 0.287***<br>(0.089)             | -0.403***<br>(0.092)            | 0.116*<br>(0.061)               | -5301***<br>(1460) | 0.121<br>(0.872)         | -0.149<br>(0.683)          |
| Branch FE                       | Yes                             | Yes                             | Yes                             | Yes                | Yes                      | Yes                        |
| Time FE                         | Yes                             | Yes                             | Yes                             | Yes                | Yes                      | Yes                        |
| 2018 Mean dep. var              | 0.185                           | 0.770                           | 0.045                           | 1684.8             | 6.837                    | 2.366                      |
| p-value                         | 0.035                           | 0.001                           | 0.213                           | 0.118              | 0.152                    | 0.152                      |
| N                               | 531                             | 531                             | 531                             | 322                | 528                      | 528                        |

*Notes:* Replication of Table 8 using a 50 km radius around historic Beta-Israel cities. The dependent variables are indicators for a city-coethnic manager (column 1), a bank-coethnic manager (column 2), and a manager coethnic with neither group (column 3); the largest loan the manager can approve without headquarters' agreement (column 4), measured in thousands of 2018 ETB and winsorized at the 95th percentile within period  $\times$  bank-city-coethnicity cells; and tenure in the bank (column 5) and branch (column 6), in years. In non-bank-coethnic cities, the three ethnicity categories in columns 1–3 are mutually exclusive and exhaustive, so their coefficients sum to zero by construction. Beta Israel Exposure is the normalized share of each bank's 2018 branch network located within 50 km of a historic Beta-Israel city, interacted with a post-2018 indicator. Panel A restricts the sample to branches in bank-coethnic cities; Panel B focuses on branches in non-bank-coethnic cities. Standard errors, clustered at the branch level, are reported in parentheses. The reported  $p$ -value tests whether the coefficient is identical across panels. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table C7: Bank-level exposure to Beta-Israel cities and lending (50 km)

| Panel A: Bank-City Coethnic     |                       |                   |                     |                     |                     |                      |
|---------------------------------|-----------------------|-------------------|---------------------|---------------------|---------------------|----------------------|
|                                 | (1)<br>Oper. distance | (2)<br>Num. loans | (3)<br>Loan size    | (4)<br>Lending rate | (5)<br>Collateral   | (6)<br>Default       |
| Beta Israel Exposure            | -3.428**<br>(1.488)   | 0.128<br>(0.319)  | 3.525***<br>(0.845) | 0.005<br>(0.005)    | 0.194***<br>(0.054) | -0.118***<br>(0.043) |
| Branch FE                       | Yes                   | Yes               | Yes                 | Yes                 | Yes                 | Yes                  |
| Time FE                         | Yes                   | Yes               | Yes                 | Yes                 | Yes                 | Yes                  |
| 2018 Mean dep. var              | 7.818                 | 2.181             | 6.041               | 0.147               | 0.986               | 0.052                |
| N                               | 308                   | 224               | 222                 | 292                 | 202                 | 273                  |
| Panel B: Not Bank-City Coethnic |                       |                   |                     |                     |                     |                      |
|                                 | Oper. distance        | Num. loans        | Loan size           | Lending rate        | Collateral          | Default              |
| Beta Israel Exposure            | 8.100***<br>(1.254)   | -0.292<br>(0.325) | 3.931***<br>(0.886) | 0.006<br>(0.005)    | -0.091**<br>(0.043) | 0.034<br>(0.045)     |
| Branch FE                       | Yes                   | Yes               | Yes                 | Yes                 | Yes                 | Yes                  |
| Time FE                         | Yes                   | Yes               | Yes                 | Yes                 | Yes                 | Yes                  |
| 2018 Mean dep. var              | 7.542                 | 1.928             | 6.255               | 0.148               | 0.995               | 0.045                |
| p-value                         | 0.000                 | 0.357             | 0.741               | 0.862               | 0.000               | 0.015                |
| N                               | 413                   | 307               | 304                 | 366                 | 267                 | 325                  |

*Notes:* Replication of Table C5 using a 50 km radius around historic Beta-Israel cities to construct the bank-level exposure measure. The dependent variables are the operational distance of the branch, the log number of outstanding loans, the log average loan size, the lending rate on a typical loan, the percentage of collateral on a typical loan, and the share of loans in default. Beta Israel Exposure is the share of each bank's branches within 50 km of a historic Beta-Israel city, normalized to lie in  $[0, 1]$  and interacted with a post-2018 dummy. Panel A restricts the sample to branches in coethnic cities; Panel B focuses on branches in non-coethnic cities. Standard errors, clustered at the branch level, are reported in parentheses. The reported  $p$ -value is for the  $t$ -test that the coefficient is identical across panels. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table C8: Bank-level exposure to Beta-Israel cities and branch managers (in-city and 50 km)

| Outcome                  | Bank and city non-coethnic |                       | Bank and city coethnic |                       |
|--------------------------|----------------------------|-----------------------|------------------------|-----------------------|
|                          | In-city                    | 50 km                 | In-city                | 50 km                 |
| City-coethnic manager    | 0.297***<br>( 0.100)       | 0.287***<br>( 0.089)  | 0.020<br>( 0.106)      | 0.020<br>( 0.090)     |
| Bank-coethnic manager    | -0.425***<br>( 0.102)      | -0.403***<br>( 0.092) | 0.020<br>( 0.106)      | 0.020<br>( 0.090)     |
| Neither-coethnic manager | 0.129*<br>( 0.072)         | 0.116*<br>( 0.061)    | -0.020<br>( 0.106)     | -0.020<br>( 0.090)    |
| Delegation               | -6005***<br>( 1610)        | -5301***<br>( 1460)   | -2174<br>( 2277)       | -1498<br>( 1944)      |
| Tenure in bank           | 0.901<br>( 1.018)          | 0.121<br>( 0.872)     | -0.736<br>( 1.239)     | -1.781*<br>( 0.998)   |
| Tenure in branch         | -0.209<br>( 0.751)         | -0.149<br>( 0.683)    | -1.325**<br>( 0.514)   | -1.319***<br>( 0.448) |

*Notes:* In-city exposure is the share of a bank's 2018 branches located in a historical Beta Israel city; 50 km exposure is the share of branches in or within 50 km of one. No other sampled city lies within 20 km, so the in-city measure is also the 20 km measure in this sample. Each period-2 share is divided by its own range, and period-1 exposure is set to zero. Regressions are estimated separately by bank-city coethnicity and include branch and time fixed effects. Parentheses contain branch-clustered standard errors. Delegation is in thousands of 2018 ETB. Stars denote 10, 5, and 1 percent significance.

## D Wild Cluster Bootstrap Inference

Our main specifications cluster standard errors at the branch level, reflecting the panel structure of the data. However, network exposure — the key independent variable — varies at the bank level, with only 16 banks in the sample. If bank-level shocks induce residual correlation across branches of the same bank, branch-level clustering may overstate statistical precision. Standard asymptotic corrections for cluster-robust standard errors perform poorly with so few clusters (A Colin Cameron, Jonah B Gelbach and Douglas L Miller, 2008).

To address this concern, we re-estimate all specifications using wild cluster bootstrap inference at the bank level. We use the `boottest` package (David Roodman, Morten Ørregaard Nielsen, James G MacKinnon and Matthew D Webb, 2019) with Webb weights and 9,999 bootstrap replications. Each table below presents two panels: Panel A reports results for branches in bank-coethnic cities (BCC), and Panel B for branches in non-bank-coethnic cities (NBCC). Both panels report the original point estimates and branch-clustered standard errors in parentheses, with bank-level bootstrap  $p$ -values in square brackets. In Panel B, the last row additionally reports the bootstrap  $p$ -value for the test that the effect of conflict differs between the two subsamples (i.e., the interaction test from equation 16). The paper’s central prediction concerns Panel B; Panel A serves as a comparison.

Table D1 reports results for the manager ethnicity regressions. All six network-exposure coefficients in non-bank-coethnic cities remain significant at the 5% level under bank-level clustering, with bootstrap  $p$ -values between 0.014 and 0.027. The differential effect between the two subsamples is significant at the 5% level or better for three of the six specifications.

Table D2 reports results for delegation. The point estimates are negative and significant within non-bank-coethnic branches using the original branch-clustered standard errors. Under bank-level wild-bootstrap inference, the corresponding  $p$ -values are 0.088, 0.168, and 0.091: the baseline and local-conflict estimates are marginally significant at 10 percent, while the city-by-time estimate is not. The bootstrap  $p$ -values for the differential relative to bank-coethnic branches are 0.459, 0.713, and 0.464.

Tables D3–D4 report results for manager reallocation and characteristics. All four reallocation outcomes are marginally significant at the 10% level. Manager tenure in the branch remains significant and tenure in the bank is marginally significant, while manager age is not. The differential effect on tenure in the branch remains significant; that on tenure in the bank does not.

Table D5 reports results for lending outcomes. Most lending outcomes lose significance, with the exception of the log number of loans ( $p = 0.017$ ), which remains significant at the 5% level. This reflects the limited statistical power of 16 bank-level clusters for detecting effects on outcomes that are plausibly noisier than the manager ethnicity outcome. The point estimates and branch-clustered standard errors are unchanged; the loss of significance reflects the more conservative inference, not a change in the estimated effects.

In summary, bank-level bootstrap inference preserves the manager-selection result and the effect on the number of loans. Delegation is marginally significant in the baseline and local-conflict specifications, imprecise with city-by-time effects, and not statistically different across the two city groups.

Table D1: Wild cluster bootstrap inference: manager ethnicity

| Panel A: Branches in Bank-Coethnic Cities     |                              |                             |                              |                              |                              |                              |
|-----------------------------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
|                                               | City coethnic manager        |                             |                              | Bank coethnic manager        |                              |                              |
|                                               | (1)                          | (2)                         | (3)                          | (4)                          | (5)                          | (6)                          |
| Network Exposure                              | -0.005<br>(0.026)<br>[0.886] | 0.070<br>(0.055)<br>[0.323] | -0.006<br>(0.026)<br>[0.863] | -0.005<br>(0.026)<br>[0.886] | 0.070<br>(0.055)<br>[0.323]  | -0.006<br>(0.026)<br>[0.863] |
| Local Exposure                                |                              |                             | 0.014<br>(0.008)<br>[.]      |                              |                              | 0.014<br>(0.008)<br>[.]      |
| Branch FE                                     | Yes                          | Yes                         | Yes                          | Yes                          | Yes                          | Yes                          |
| Time FE                                       | Yes                          | No                          | Yes                          | Yes                          | No                           | Yes                          |
| City-Time FE                                  | No                           | Yes                         | No                           | No                           | Yes                          | No                           |
| 2018 Mean dep. var                            | 0.680                        | 0.695                       | 0.680                        | 0.680                        | 0.695                        | 0.680                        |
| N (Branches)                                  | 447                          | 371                         | 447                          | 447                          | 371                          | 447                          |
| Panel B: Branches in Non-Bank-Coethnic Cities |                              |                             |                              |                              |                              |                              |
|                                               | City coethnic manager        |                             |                              | Bank coethnic manager        |                              |                              |
|                                               | (1)                          | (2)                         | (3)                          | (4)                          | (5)                          | (6)                          |
| Network Exposure                              | 0.050<br>(0.016)<br>[0.025]  | 0.047<br>(0.017)<br>[0.019] | 0.051<br>(0.016)<br>[0.027]  | -0.097<br>(0.015)<br>[0.014] | -0.096<br>(0.017)<br>[0.021] | -0.099<br>(0.015)<br>[0.015] |
| Local Exposure                                |                              |                             | -0.014<br>(0.009)<br>[.]     |                              |                              | 0.031<br>(0.010)<br>[.]      |
| Branch FE                                     | Yes                          | Yes                         | Yes                          | Yes                          | Yes                          | Yes                          |
| Time FE                                       | Yes                          | No                          | Yes                          | Yes                          | No                           | Yes                          |
| City-Time FE                                  | No                           | Yes                         | No                           | No                           | Yes                          | No                           |
| 2018 Mean dep. var                            | 0.185                        | 0.181                       | 0.185                        | 0.770                        | 0.769                        | 0.770                        |
| p-value diff vs BCC (bootstrap)               | 0.258                        | 0.834                       | 0.252                        | 0.006                        | 0.050                        | 0.001                        |
| N (Branches)                                  | 531                          | 464                         | 531                          | 531                          | 464                          | 531                          |

*Notes:* This table replicates Table 2. Panel A restricts the sample to branches in cities coethnic with the bank (BCC); Panel B focuses on branches in cities that are not (NBCC). Point estimates and branch-clustered standard errors (in parentheses) are identical to the main table. Bank-level wild cluster bootstrap  $p$ -values for Network Exposure are reported in square brackets, computed using Webb weights with 9,999 replications and clustering at the bank level (16 banks); the bootstrap  $p$ -value for the local-exposure control is not reported. The last row of Panel B reports the bootstrap  $p$ -value for the null that the effect of network exposure is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table D2: Wild cluster bootstrap inference: delegation

| Panel A: Branches in Bank-Coethnic Cities     |                              |                               |                              |
|-----------------------------------------------|------------------------------|-------------------------------|------------------------------|
|                                               | (1)                          | (2)                           | (3)                          |
| Network Exposure                              | -111.9<br>(381.9)<br>[0.861] | -1178.3<br>(839.1)<br>[0.406] | -115.6<br>(384.5)<br>[0.855] |
| Local Exposure                                |                              |                               | -422.3<br>(172.1)<br>[.]     |
| Branch FE                                     | Yes                          | Yes                           | Yes                          |
| Time FE                                       | Yes                          | No                            | Yes                          |
| City-Time FE                                  | No                           | Yes                           | No                           |
| 2018 Mean dep. var                            | 1852.8                       | 1993.8                        | 1852.8                       |
| N (Branches)                                  | 231                          | 160                           | 231                          |
| Panel B: Branches in Non-Bank-Coethnic Cities |                              |                               |                              |
|                                               | (1)                          | (2)                           | (3)                          |
| Network Exposure                              | -633.1<br>(282.0)<br>[0.088] | -666.1<br>(330.3)<br>[0.168]  | -627.0<br>(281.5)<br>[0.091] |
| Local Exposure                                |                              |                               | -60.7<br>(135.6)<br>[.]      |
| Branch FE                                     | Yes                          | Yes                           | Yes                          |
| Time FE                                       | Yes                          | No                            | Yes                          |
| City-Time FE                                  | No                           | Yes                           | No                           |
| 2018 Mean dep. var                            | 1684.8                       | 1721.8                        | 1684.8                       |
| p-value diff vs BCC (bootstrap)               | 0.459                        | 0.713                         | 0.464                        |
| N (Branches)                                  | 322                          | 266                           | 322                          |

*Notes:* This table replicates Table 3. The dependent variable is the largest loan a branch manager can approve without headquarters' agreement, measured in thousands of 2018 ETB and winsorized at the 95th percentile. Panel A restricts the sample to branches in cities coethnic with the bank (BCC); Panel B focuses on branches in cities that are not (NBCC). Point estimates and branch-clustered standard errors (in parentheses) are identical to the main table. Bank-level wild cluster bootstrap  $p$ -values for Network Exposure are reported in square brackets; the bootstrap  $p$ -value for the local-exposure control is not reported. The last row of Panel B reports the bootstrap  $p$ -value for the null that the effect is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table D3: Wild cluster bootstrap inference: manager reallocation

| Panel A: Branches in Bank-Coethnic Cities     |                              |                              |                              |                              |
|-----------------------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
|                                               | New manager<br>bank          | New manager<br>branch        | Reallocated                  | New hire                     |
|                                               | (1)                          | (2)                          | (3)                          | (4)                          |
| Network Exposure                              | 0.016<br>(0.009)<br>[0.234]  | -0.015<br>(0.032)<br>[0.618] | -0.027<br>(0.031)<br>[0.278] | 0.011<br>(0.008)<br>[0.307]  |
| Mean dep. var                                 | 0.034                        | 0.397                        | 0.370                        | 0.027                        |
| N (Branches)                                  | 447                          | 446                          | 446                          | 446                          |
| Panel B: Branches in Non-Bank-Coethnic Cities |                              |                              |                              |                              |
|                                               | New manager<br>bank          | New manager<br>branch        | Reallocated                  | New hire                     |
|                                               | (1)                          | (2)                          | (3)                          | (4)                          |
| Network Exposure                              | -0.010<br>(0.008)<br>[0.059] | 0.065<br>(0.024)<br>[0.083]  | 0.075<br>(0.024)<br>[0.084]  | -0.010<br>(0.008)<br>[0.059] |
| Mean dep. var                                 | 0.017                        | 0.360                        | 0.345                        | 0.015                        |
| p-value diff vs BCC (bootstrap)               | 0.141                        | 0.387                        | 0.094                        | 0.151                        |
| N (Branches)                                  | 528                          | 528                          | 528                          | 528                          |

*Notes:* This table replicates Table 4. The sample is restricted to the 2022 cross-section. The four outcomes are whether the manager joined the bank (column 1) or branch (column 2) after the onset of conflict, and conditional on joining the branch, whether they were reallocated (column 3) or a new hire (column 4). Panel A restricts the sample to branches in cities coethnic with the bank (BCC); Panel B focuses on branches in cities that are not (NBCC). Point estimates and branch-clustered standard errors (in parentheses) are identical to the main table. Bank-level wild cluster bootstrap  $p$ -values are reported in square brackets. The last row of Panel B reports the bootstrap  $p$ -value for the null that the effect is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table D4: Wild cluster bootstrap inference: manager characteristics

| Panel A: Branches in Bank-Coethnic Cities     |                             |                              |                              |                              |                             |
|-----------------------------------------------|-----------------------------|------------------------------|------------------------------|------------------------------|-----------------------------|
|                                               | Tenure in<br>bank           | Tenure in<br>branch          | Age                          | University<br>degree         | Male<br>manager             |
|                                               | (1)                         | (2)                          | (3)                          | (4)                          | (5)                         |
| Network Exposure                              | 0.029<br>(0.230)<br>[0.907] | -0.265<br>(0.126)<br>[0.019] | -0.026<br>(0.354)<br>[0.957] | -0.002<br>(0.002)<br>[0.444] | 0.008<br>(0.028)<br>[0.897] |
| Branch FE                                     | Yes                         | Yes                          | Yes                          | Yes                          | Yes                         |
| Time FE                                       | Yes                         | Yes                          | Yes                          | Yes                          | Yes                         |
| 2018 Mean dep. var                            | 6.859                       | 2.390                        | 33.825                       | 0.993                        | 0.868                       |
| N (Branches)                                  | 447                         | 446                          | 417                          | 448                          | 448                         |
| Panel B: Branches in Non-Bank-Coethnic Cities |                             |                              |                              |                              |                             |
|                                               | Tenure in<br>bank           | Tenure in<br>branch          | Age                          | University<br>degree         | Male<br>manager             |
|                                               | (1)                         | (2)                          | (3)                          | (4)                          | (5)                         |
| Network Exposure                              | 0.498<br>(0.154)<br>[0.074] | 0.221<br>(0.098)<br>[0.018]  | 0.538<br>(0.202)<br>[0.116]  | -0.002<br>(0.001)<br>[0.497] | 0.008<br>(0.014)<br>[0.560] |
| Branch FE                                     | Yes                         | Yes                          | Yes                          | Yes                          | Yes                         |
| Time FE                                       | Yes                         | Yes                          | Yes                          | Yes                          | Yes                         |
| 2018 Mean dep. var                            | 6.837                       | 2.366                        | 33.700                       | 0.989                        | 0.866                       |
| p-value diff vs BCC (bootstrap)               | 0.116                       | 0.004                        | 0.257                        | 0.952                        | 0.998                       |
| N (Branches)                                  | 528                         | 528                          | 500                          | 529                          | 529                         |

*Notes:* This table replicates Table 5. The dependent variables are the manager's tenure in the bank, tenure in the current branch, age, whether they hold a university degree, and gender. Panel A restricts the sample to branches in cities coethnic with the bank (BCC); Panel B focuses on branches in cities that are not (NBCC). Point estimates and branch-clustered standard errors (in parentheses) are identical to the main table. Bank-level wild cluster bootstrap  $p$ -values are reported in square brackets. The last row of Panel B reports the bootstrap  $p$ -value for the null that the effect is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table D5: Wild cluster bootstrap inference: lending outcomes

| Panel A: Branches in Bank-Coethnic Cities     |                              |                              |                             |                              |                              |                              |
|-----------------------------------------------|------------------------------|------------------------------|-----------------------------|------------------------------|------------------------------|------------------------------|
|                                               | Operational distance         | Log number of loans          | Log average loan size       | Average lending rate         | Collateral percentage        | Share loan default           |
|                                               | (1)                          | (2)                          | (3)                         | (4)                          | (5)                          | (6)                          |
| Network Exposure                              | -1.658<br>(0.681)<br>[0.006] | 0.049<br>(0.117)<br>[0.591]  | 0.395<br>(0.158)<br>[0.244] | 0.001<br>(0.001)<br>[0.528]  | 0.015<br>(0.013)<br>[0.663]  | -0.026<br>(0.017)<br>[0.136] |
| Branch FE                                     | Yes                          | Yes                          | Yes                         | Yes                          | Yes                          | Yes                          |
| Time FE                                       | Yes                          | Yes                          | Yes                         | Yes                          | Yes                          | Yes                          |
| 2018 Mean dep. var                            | 7.818                        | 2.181                        | 6.041                       | 0.147                        | 0.986                        | 0.052                        |
| N (Branches)                                  | 308                          | 224                          | 222                         | 292                          | 202                          | 273                          |
| Panel B: Branches in Non-Bank-Coethnic Cities |                              |                              |                             |                              |                              |                              |
|                                               | Operational distance         | Log number of loans          | Log average loan size       | Average lending rate         | Collateral percentage        | Share loan default           |
|                                               | (1)                          | (2)                          | (3)                         | (4)                          | (5)                          | (6)                          |
| Network Exposure                              | -0.084<br>(0.157)<br>[0.832] | -0.233<br>(0.051)<br>[0.017] | 0.123<br>(0.133)<br>[0.584] | -0.000<br>(0.001)<br>[0.958] | -0.006<br>(0.008)<br>[0.526] | -0.007<br>(0.007)<br>[0.444] |
| Branch FE                                     | Yes                          | Yes                          | Yes                         | Yes                          | Yes                          | Yes                          |
| Time FE                                       | Yes                          | Yes                          | Yes                         | Yes                          | Yes                          | Yes                          |
| 2018 Mean dep. var                            | 7.542                        | 1.928                        | 6.255                       | 0.148                        | 0.995                        | 0.045                        |
| p-value diff vs BCC (bootstrap)               | 0.018                        | 0.001                        | 0.488                       | 0.638                        | 0.512                        | 0.152                        |
| N (Branches)                                  | 413                          | 307                          | 304                         | 366                          | 267                          | 325                          |

*Notes:* This table replicates Table 6. The dependent variables are the operational distance, log number of loans, log average loan size, average lending rate, collateral percentage, and share of loans in default. Panel A restricts the sample to branches in cities coethnic with the bank (BCC); Panel B focuses on branches in cities that are not (NBCC). Point estimates and branch-clustered standard errors (in parentheses) are identical to the main table. Bank-level wild cluster bootstrap  $p$ -values are reported in square brackets. The last row of Panel B reports the bootstrap  $p$ -value for the null that the effect is identical in bank-coethnic and non-bank-coethnic cities. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

## E Ethnicity of banks, cities and managers

### E.1 Comments on specific choices made in the analysis

**Bank ethnicity.** We classify each of the 16 commercial banks in our sample into a single dominant ethnicity using information from the 2018 annual reports. For each bank, we collect the names of all members of the board of directors and identify each director’s ethnic group from their name, following the conventional correspondence between Ethiopian naming conventions and ethnic affiliation. A bank is assigned the plurality ethnicity of its board. The two state-owned banks, the Commercial Bank of Ethiopia and the Construction and Business Bank, are classified as Amhara; the latter was effectively absorbed into the former during our sample period. This procedure yields 9 Amhara-dominated banks, 3 Oromo, 2 SNNP, and 2 Tigray.

**City ethnicity.** We assign each city its historical ethnic homeland using the digitized ethnic-boundary shapefile from [Nathan Nunn \(2008\)](#), which encodes the ethnic territories of [George Peter Murdock \(1959\)](#). We spatially match each branch’s geographic coordinates to a Murdock polygon using the `geoinpoly` routine. Murdock’s tribe labels are pre-modern ethnographic categories (e.g., “Galla,” “Somal”); we map these to the contemporary ethnic categories used in our analysis (Oromo, Amhara, Tigray, Somali, SNNP, Beta Israel, Other) using a correspondence table constructed from Wikipedia entries for each Murdock tribe. The full mapping is available in the replication package (`1rawData/13EthnicGroups/tribeEthnicityMapping.xlsx`). Addis Ababa, which lies on the boundary between Oromo and Amhara territories in Murdock’s classification, is assigned to the Amhara category, reflecting the dominant ethnic composition of the city’s banking sector and civil service during our sample period.

**Treatment of the Harari.** The Harari are a small ethnic group (approximately 0.2% of the Ethiopian population in the 2007 census) whose traditional homeland is the walled city of Harar and its immediate surroundings, located within predominantly Oromo territory. No manager in our survey sample self-identifies as Harari, so Harari does not appear as a category in the manager-ethnicity classification. For the city-ethnicity assignment based on the [Nunn \(2008\)](#) digitization of the Murdock Atlas, Harar and its surrounding cities fall within the Oromo polygon and are therefore assigned to the Oromo category. This coding has no material effect on our results: only 7 branches in our panel are located in cities of the historic Harari homeland.

**Aggregation of SNNP groups.** The Southern Nations, Nationalities, and Peoples (SNNP) region of Ethiopia is home to more than 50 officially recognized ethnic groups, eight of which appear in our sample: Sidama, Gurage, Gamo, Wolayta, Hadiya, Kambaata, Kafa, and Silte. Because bank and city ethnicity in our main specifications are defined at a coarser level (the bank’s dominant ethnicity and the city’s Murdock-Atlas classification), we aggregate these eight groups into a single “SNNP” category for the coethnicity indicators. This follows standard practice in the Ethiopian ethnic-politics literature and is also consistent with the self-identification reported by several managers in our data.

### E.2 List of banks and bank-ethnicities

Table E1: Distribution of banks by ethnic affiliation

| Bank name           | Bank ethnicity | Count | Share (%) |
|---------------------|----------------|-------|-----------|
| Abay                | Amhara         | 80    | 8.2       |
| Addis               | Amhara         | 5     | 0.5       |
| Awash               | Oromo          | 100   | 10.2      |
| BOA                 | Amhara         | 37    | 3.8       |
| Berhan              | Amhara         | 72    | 7.4       |
| Bunna International | Amhara         | 46    | 4.7       |
| CBB                 | Amhara         | 19    | 1.9       |
| CBE                 | Amhara         | 210   | 21.5      |
| CBO                 | Oromo          | 53    | 5.4       |
| Dashen              | Amhara         | 55    | 5.6       |
| Debub               | SNNP           | 17    | 1.7       |
| Lion                | Tigray         | 14    | 1.4       |
| NIB International   | SNNP           | 86    | 8.8       |
| OIB                 | Oromo          | 75    | 7.7       |
| United              | Amhara         | 51    | 5.2       |
| Wegagen             | Tigray         | 59    | 6.0       |

*Notes:* This table reports the ethnic affiliation of each bank in the sample, the number of surveyed branches, and the corresponding share of the total sample. Bank ethnicity is the dominant ethnic affiliation of the bank's leadership.

### E.3 Number of cities by ethnic group

Table E2: Distribution of cities by ethnic group

| City ethnicity    | Count | Share (%) |
|-------------------|-------|-----------|
| Afar              | 4     | 0.4       |
| Amhara            | 522   | 53.3      |
| Anuak             | 4     | 0.4       |
| Benishangul-Gumuz | 2     | 0.2       |
| Oromo             | 260   | 26.6      |
| SNNP              | 160   | 16.3      |
| Somali            | 17    | 1.7       |
| Tigray            | 10    | 1.0       |

*Notes:* This table reports the distribution of branch locations by ethnic group. City ethnicity is determined using the Murdock Ethnographic Atlas. Count and share refer to the number and percentage of branches located in cities of each ethnic group.

## E.4 Number of managers by ethnic group by period

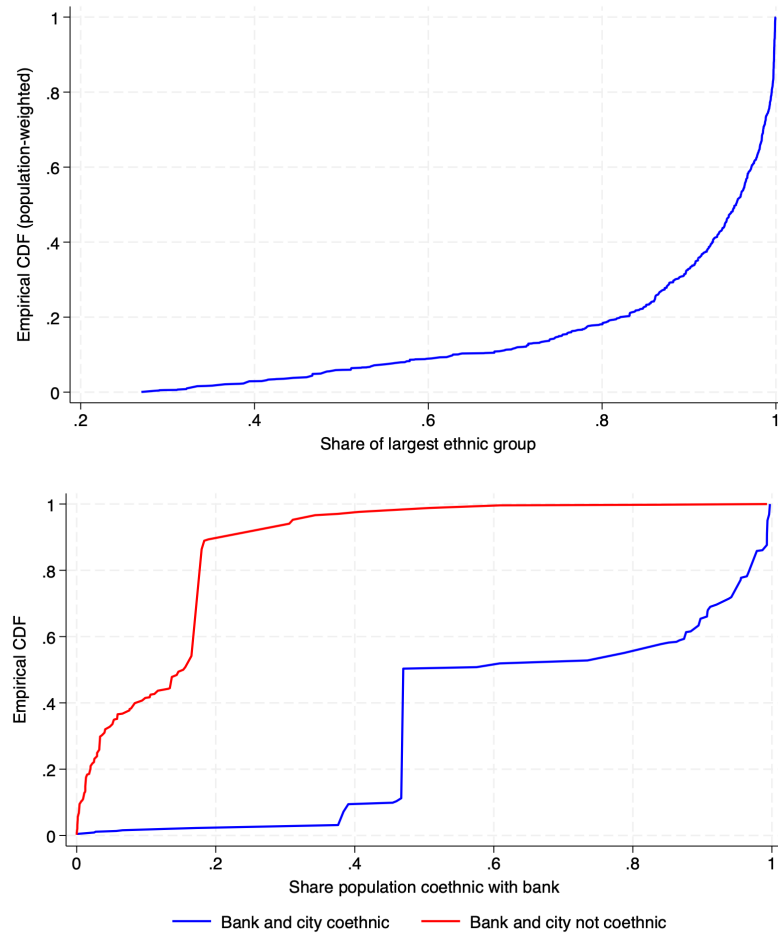
Table E3: Distribution of branch managers' ethnicity by period

| Manager ethnicity | First Wave (2018) |           | Second Wave (2022) |           |
|-------------------|-------------------|-----------|--------------------|-----------|
|                   | Count             | Share (%) | Count              | Share (%) |
| Afar              | 1                 | 0.1       | 5                  | 0.5       |
| Agew              | 0                 | 0.0       | 8                  | 0.8       |
| Amhara            | 495               | 50.6      | 433                | 44.2      |
| Benishangul-Gumuz | 2                 | 0.2       | 1                  | 0.1       |
| Oromo             | 289               | 29.5      | 380                | 38.8      |
| SNNP              | 109               | 11.1      | 112                | 11.4      |
| Somali            | 2                 | 0.2       | 19                 | 1.9       |
| Tigray            | 81                | 8.3       | 20                 | 2.0       |

*Notes:* This table reports the distribution of branch managers' ethnicity across survey waves. Manager ethnicity is identified using multiple indicators including region of birth, accent, spoken languages, and name, determined jointly by two interviewers.

## E.5 Ethnic diversity at the local level

Figure E1: Ethnic diversity in Ethiopia



*Notes:* This figure shows, in the top panel, the empirical CDF for the share of the population that belongs to largest ethnic group in each woreda – the smallest administrative unit – in Ethiopia. It shows that in over 80% of woredas over 80% of people belong to a single ethnicity. The bottom panel instead shows that share of the population in each zone (region in the case of Addis Ababa) in which a branch is located that belongs to the same ethnic group of the bank. The differences in administrative units used in this figure is based on how (im)precisely we observe the location of bank branches. This is based on 2007 IPUMS data and corresponding administrative boundaries.

## F Management Practices Survey and Methodology

### F.1 Scoring Management Performances

The concept of “good management practice” is often relative and contingent to a specific market or sector. However, international best practices in the banking sector have been widely studied, allowing the literature to reach a consensus on what constitutes a “good” practice. We evaluate and score management practices by defining the concept of “good” and “bad” practice and codifying it from 1 (worst practice) to 5 (best practice) across eight key dimensions. These practices are grouped into four areas: lean management, performance management, target management, and people management.<sup>1</sup>

The lean management section evaluates the efficiency of day-to-day operations (e.g., organization of the workflow, management of slack time, method of staff assignment to specific tasks, etc.) and assesses whether the branch has adopted recognized best practices. The performance management part measures how performance is tracked and reviewed (e.g., what KPIs are used, who oversees the performance review, how frequently the performance measures are revised, etc.). The target management section reviews the nature and scope of a branch’s targets (e.g., the relative importance of financial targets compared to non-financial ones, how these targets are linked to performance measures, etc.) and assesses whether they align with the bank’s objectives. Finally, the people management part tests whether good performance is appropriately rewarded and whether bad performance is sanctioned (e.g., adoption of an articulated appraisal system, use of a reward plan, presence of sanctions for underperformers, etc.).

An overview of these four sections is presented in Table F1. After the completion of the survey, we combine these scores to obtain a unique index of management practices. Since the scaling may vary across practices in the econometric estimation, we normalize the final scores to  $z$ -scores (i.e., mean zero and standard deviation one). To build our final management index, we take the unweighted average across all  $z$ -scores as

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<sup>1</sup> Note, this section is taken directly from a previously circulated working paper (Nicola Gennaioli, Nicola Limodio and Francesco Strobbe, 2019)

our primary measure of overall management practice.

Table F1: Survey Indicators

| (1) Indicator                   | (2) Objective                             | (3) Measure                                                                                                         |
|---------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>1 Lean Management</b>        |                                           |                                                                                                                     |
| 1.1                             | Introduction of new management techniques | Tests whether operational efficiencies have been introduced and why                                                 |
| 1.2                             | Good use of human resources               | Tests whether incoming demand is segmented appropriately and the efficiency at matching supply and demand of skills |
| <b>2 Performance Management</b> |                                           |                                                                                                                     |
| 2.1                             | Performance tracking                      | Tests whether performance is tracked using meaningful metrics and with appropriate regularity                       |
| 2.2                             | Review of performance                     | Tests whether performance is reviewed with appropriate frequency and communicated with staff                        |
| <b>3 Target Management</b>      |                                           |                                                                                                                     |
| 3.1                             | Target balance                            | Tests whether targets cover a sufficiently broad set of metrics                                                     |
| 3.2                             | Target interconnections                   | Tests whether targets are tied to company objectives and how well they cascade down the organization                |
| <b>4 People Management</b>      |                                           |                                                                                                                     |
| 4.1                             | Building a high performance culture       | Tests whether good performance is rewarded proportionately                                                          |
| 4.2                             | Making room for talent                    | Tests whether the branch is able to deal with underperformers                                                       |

*Notes:* This table illustrates our survey's section on management performance. To measure good managerial practices, we evaluate them along four main categories, i.e., lean management, performance management, target management, and people management. Each category is scored on two dimensions on a scale from 1 (bad practice) to 5 (good practice). Our final management performance index is calculated by standardizing each of the eight resulting scores and by taking their unweighted average. Each of the sub-indices is calculated in the same way using a combination of two metrics (respectively, the lean, performance, target and people management practices).

## **F.2 Responses Collection**

The information content and precision of our management performance index crucially depend on the quality of the responses collected through our survey. For this reason, we structured our questions to allow respondents to provide accurate and unbiased answers. As pointed out by (Marianne Bertrand and Sendhil Mullainathan, 2001), responses are typically biased by the presence of a scoring scale, which may prompt respondents to provide the interviewer's expected answer. Additionally, interviewers may have preconceptions toward the interviewed managers. Apart from these issues, many background factors correlated with management behavior may systematically bias survey data.

To counter these potential negative effects, we adopt a combination of strategies tested in the existing literature. First, we ensure accurate responses by conducting telephone surveys without informing managers that their answers will be evaluated against a scoring grid, allowing us to gather information about actual management practices. Second, we ask a series of open-ended questions (e.g., "What types of targets are set for the bank in general? What are the goals for your branch?") and record responses until an accurate assessment of management practices is possible. Third, to correct any inconsistent interpretation of responses, we ensure interviewers conduct a minimum number of training interviews during a pilot survey phase. Furthermore, since almost all our interviewers conducted over 40 interviews (with an average of 200 interviews per interviewer), we are able to control for interviewer fixed effects in a robustness section. Fourth, we adopt a double-scoring technique, where another interviewer silently listens and scores the responses provided during the interview. Finally, we collect information on a large set of manager characteristics (e.g., age, education, ethnicity, etc.) to control for potential confounders.

## **F.3 Manager Participation**

The average duration of an interview was approximately 27 minutes. Interviews were conducted between January 2015 and February 2018 by locally recruited research assistants (RAs). The initial survey phase consisted of a pilot involving 265 branches, which allowed us to refine the survey and train interviewers. After this stage, we surveyed the universe of Ethiopian bank branches (3,232 in total). We achieved a relatively high response rate (59%, 1,910 branches) through specific strategies. First, the survey was introduced as a neutral exercise, avoiding discussion of the bank's financial position or individual branch accounts. This ensured that interviewers were blind to financial data and helped maximize participation. Second, questions focused on evident practices within the branch, enabling branch managers to provide reliable answers. Finally, endorsement and support from the World Bank and the National Bank of Ethiopia encouraged participation by emphasizing the initiative's importance and official backing.

## **F.4 Additional Data Collected**

In addition to management practices, our survey collected extensive data on respondent and branch characteristics. The introductory section gathered baseline information about the branch (e.g., name, geographic location, number of employees, opening date) and the manager (e.g., age, gender, education, previous employment).

A second set of questions addressed the organizational nature of the branch (e.g., management structure, autonomy from headquarters) and operations (e.g., average loan size, collateral requirements, share of

defaulted loans).

## **F.5 Summary statistics by wave**

Tables [F2](#) and [F3](#) report summary statistics on the main analysis variables by survey wave for respectively organizational characteristics and characteristics related to lending activities.

Table F2: Summary statistics of organizational variables by wave

| <i>Panel A: Manager Characteristics</i> |       |          |           |        |           |
|-----------------------------------------|-------|----------|-----------|--------|-----------|
|                                         | N     | Mean     | Std. Dev. | Min    | Max       |
| Bank coethnic manager                   | 1,957 | 0.65     | 0.48      | 0      | 1         |
| First Wave (2018)                       | 979   | 0.73     | 0.45      | 0      | 1         |
| Second Wave (2022)                      | 978   | 0.57     | 0.49      | 0      | 1         |
| City coethnic manager                   | 1,957 | 0.54     | 0.50      | 0      | 1         |
| First Wave (2018)                       | 979   | 0.41     | 0.49      | 0      | 1         |
| Second Wave (2022)                      | 978   | 0.66     | 0.47      | 0      | 1         |
| Manager born in region                  | 1,439 | 0.42     | 0.49      | 0      | 1         |
| First Wave (2018)                       | 464   | 0.44     | 0.50      | 0      | 1         |
| Second Wave (2022)                      | 975   | 0.42     | 0.49      | 0      | 1         |
| Tenure in bank (years)                  | 1,954 | 7.71     | 4.44      | 0      | 36        |
| First Wave (2018)                       | 979   | 6.85     | 4.27      | 0      | 35        |
| Second Wave (2022)                      | 975   | 8.57     | 4.45      | 0      | 36        |
| Tenure in branch (years)                | 1,953 | 2.67     | 2.62      | 0      | 35        |
| First Wave (2018)                       | 979   | 2.37     | 2.33      | 1      | 35        |
| Second Wave (2022)                      | 974   | 2.98     | 2.85      | 0      | 23        |
| Age                                     | 1,896 | 34.00    | 5.67      | 16     | 60        |
| First Wave (2018)                       | 979   | 33.70    | 6.02      | 16     | 60        |
| Second Wave (2022)                      | 917   | 34.32    | 5.25      | 23     | 56        |
| University degree                       | 1,956 | 0.99     | 0.07      | 0      | 1         |
| First Wave (2018)                       | 979   | 0.99     | 0.10      | 0      | 1         |
| Second Wave (2022)                      | 977   | 1.00     | 0.00      | 1      | 1         |
| Male manager                            | 1,956 | 0.83     | 0.38      | 0      | 1         |
| First Wave (2018)                       | 979   | 0.87     | 0.34      | 0      | 1         |
| Second Wave (2022)                      | 977   | 0.79     | 0.40      | 0      | 1         |
| <i>Panel B: Branch Characteristics</i>  |       |          |           |        |           |
|                                         | N     | Mean     | Std. Dev. | Min    | Max       |
| Share local employees                   | 1,920 | 0.57     | 0.25      | 0.10   | 1.00      |
| First Wave (2018)                       | 979   | 0.57     | 0.24      | 0.10   | 1.00      |
| Second Wave (2022)                      | 941   | 0.57     | 0.25      | 0.10   | 1.00      |
| Branch employees                        | 1,951 | 17.57    | 10.04     | 3      | 110       |
| First Wave (2018)                       | 979   | 14.77    | 10.32     | 3      | 110       |
| Second Wave (2022)                      | 972   | 20.40    | 8.90      | 5      | 78        |
| Max loan w/o HQ                         | 1,532 | 3,854.91 | 6,079.66  | 0.00   | 29,823.23 |
| First Wave (2018)                       | 979   | 1,681.82 | 1,614.30  | 0.00   | 5,500.00  |
| Second Wave (2022)                      | 553   | 7,702.04 | 8,642.75  | 149.12 | 29,823.23 |

*Notes:* This table reports summary statistics of organisational analysis variables by survey wave, organised into two panels. Panel A reports manager characteristics: bank and city coethnic manager indicate whether the manager shares the ethnicity of the bank or city, respectively; manager born in region indicates whether the manager was born in the same region as the branch; tenure in bank and branch are measured in years; university degree and male manager are dummy indicators. Panel B reports branch characteristics: share local employees is the share of branch employees from the local area; branch employees is the headcount; max loan w/o HQ is the largest loan a branch manager can approve without headquarters' agreement, in ETB 1,000, deflated to 2018 prices using annual CPI inflation rates (World Bank WDI, FP.CPI.TOTL.ZG) and winsorized at the 95th percentile within period  $\times$  bank-city coethnicity cells.

Table F3: Summary statistics of lending variables by wave

|                            | N     | Mean | Std. Dev. | Min   | Max   |
|----------------------------|-------|------|-----------|-------|-------|
| Operational distance (km)  | 1,700 | 6.80 | 4.85      | 0.05  | 50.00 |
| First Wave (2018)          | 979   | 7.62 | 2.62      | 3.00  | 30.00 |
| Second Wave (2022)         | 721   | 5.70 | 6.64      | 0.05  | 50.00 |
| Log number of loans        | 1,510 | 2.43 | 1.34      | 0.00  | 8.52  |
| First Wave (2018)          | 979   | 2.05 | 1.27      | 0.00  | 6.01  |
| Second Wave (2022)         | 531   | 3.13 | 1.18      | 0.00  | 8.52  |
| Log average loan size      | 1,505 | 6.30 | 3.13      | -4.61 | 10.95 |
| First Wave (2018)          | 979   | 6.05 | 3.83      | -4.61 | 10.95 |
| Second Wave (2022)         | 526   | 6.77 | 0.65      | 5.70  | 8.51  |
| Average lending rate       | 1,637 | 0.15 | 0.02      | 0.10  | 0.20  |
| First Wave (2018)          | 979   | 0.15 | 0.02      | 0.10  | 0.20  |
| Second Wave (2022)         | 658   | 0.15 | 0.02      | 0.10  | 0.20  |
| Collateral (share of loan) | 1,448 | 0.95 | 0.19      | 0.50  | 1.50  |
| First Wave (2018)          | 979   | 1.00 | 0.20      | 0.50  | 1.50  |
| Second Wave (2022)         | 469   | 0.83 | 0.13      | 0.50  | 1.00  |
| Share loan default         | 1,577 | 0.07 | 0.14      | 0.00  | 1.00  |
| First Wave (2018)          | 979   | 0.05 | 0.09      | 0.00  | 1.00  |
| Second Wave (2022)         | 598   | 0.09 | 0.20      | 0.00  | 1.00  |

*Notes:* This table reports summary statistics of lending analysis variables by survey wave. Operational distance is the maximum distance (in km) at which a branch services customers. Log number of loans and log average loan size are in natural logarithms, with loan size expressed in ETB 1,000. Average lending rate is the mean rate on the past 10 loans. Collateral is the typical collateral as a share of total loan size. Share loan default is the share of defaulting loans in the past year.

## G Ethnic conflict over time and space

Figure G1: Regions of Ethiopia



*Notes:* This figure displays the twelve regions in Ethiopia and two administrative councils, Addis Ababa and Dire Dawa. To ensure regions stay the same over time in our analysis, we combine Central Ethiopia, South West Ethiopia, South Ethiopia and Sidama into the Southern Nations, Nationalities and Peoples' Region (SNNPR). This region was split into four between 2020 and 2023, and remains one of the most ethnically diverse regions of the country.

### G.1 Mapping ACLED actors to ethnic groups

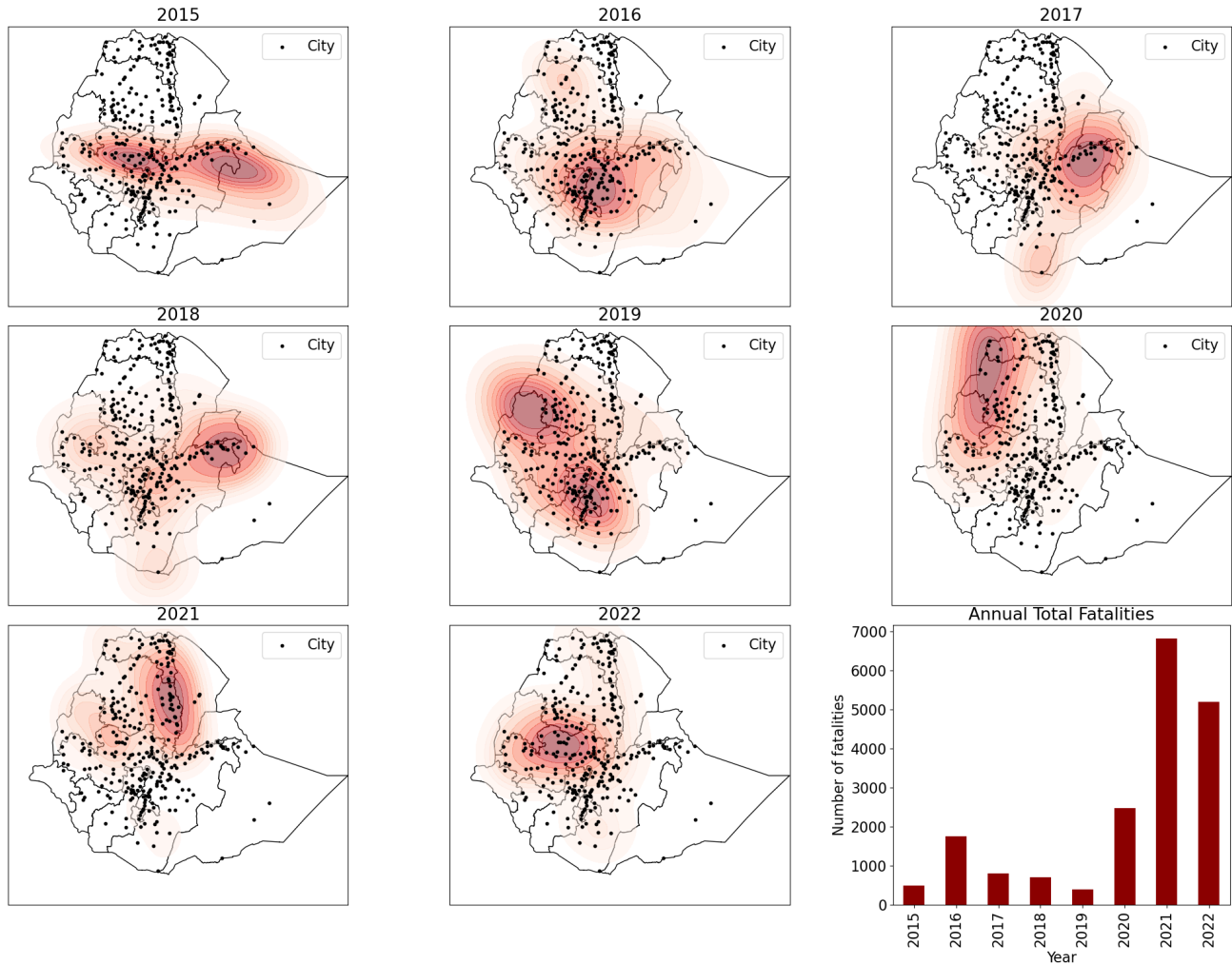
We identify the ethnic groups involved in each ACLED event from the names in the two primary actor fields and all associated-actor fields. The crosswalk combines literal ethnic identifiers in ACLED's actor labels with classifications from the Ethiopia Peace Observatory and the regional concordance used throughout the analysis. Table G1 reports every substring used in the crosswalk and the corresponding assigned group. The group indicators are non-exclusive because the exposure measure records whether an event involves a bank's ethnic group. If an event involves multiple groups, its fatalities enter the exposure measure once for each involved group, and never more than once within the same group.

Table G1: ACLED actor-to-ethnicity crosswalk

| Assigned group                                 | Exact case-sensitive actor-name substrings                                                                                                                                                                                          | Classification basis                                                                                                                                             | Fatal events |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <i>Panel A: Ethnic-group crosswalk</i>         |                                                                                                                                                                                                                                     |                                                                                                                                                                  |              |
| Oromo                                          | Oromo; Oromia; Omoro; Qeerroo                                                                                                                                                                                                       | Literal ethnic and regional identifiers; the EPO actor profile classifies Qeerroo as an Oromo youth movement.                                                    | 1,263        |
| Tigray                                         | Tigray; Samri; Welkait                                                                                                                                                                                                              | Literal ethnic identifiers; the EPO actor profile classifies Samri as a Tigrayan youth group; Welkait follows the project crosswalk.                             | 443          |
| Amhara                                         | Amhara; Fano                                                                                                                                                                                                                        | Literal ethnic identifiers; the EPO actor profile classifies Fano as an Amhara militia.                                                                          | 422          |
| Somali                                         | Somali; Garre; Gabra                                                                                                                                                                                                                | Literal ethnic identifiers and the project clan crosswalk. Actor labels referring to Somalia rather than Ethiopian Somali groups are excluded.                   | 79           |
| Ogaden                                         | Ogaden                                                                                                                                                                                                                              | Literal clan and organization identifiers, including the Ogaden National Liberation Front; retained as a separate category.                                      | 49           |
| Afar                                           | Afar                                                                                                                                                                                                                                | Literal ethnic and regional identifiers.                                                                                                                         | 103          |
| Gumuz                                          | Gumuz                                                                                                                                                                                                                               | Literal ethnic identifiers.                                                                                                                                      | 60           |
| SNNP aggregate                                 | Amaro; Konso; Guji; Sidama; Derashe; Dirashe; Wolayta; Wolayita; Welayta; Gurage; Gamo; Gofa; Silte; Burji; Kebena; Mareko; Ale; Xonsita; Mashole; Moseye; Kefa; Dawero; Sheka; Konta; Dorze; Gedeo; Koore; Meskan; Hadiya; Shinash | Groups combined in the project's stable SNNP category using the EPO SNNPR regional profile and the regional concordance used for bank and city ethnicity.        | 60           |
| <i>Panel B: Event-level coding conventions</i> |                                                                                                                                                                                                                                     |                                                                                                                                                                  |              |
| Actor fields                                   | actor1; actor2; all semicolon-delimited<br>assoc_actor_1 and assoc_actor_2 entries                                                                                                                                                  | A group is coded as involved when any primary or associated actor contains one of its Panel A strings.                                                           | 2,705 total  |
| Multiple groups                                | Retain every group matched within the same event                                                                                                                                                                                    | The indicators are non-exclusive: fatalities from a multi-group event enter each involved group's exposure once, and never more than once within a group.        | 472          |
| Other actors                                   | No Panel A string match                                                                                                                                                                                                             | Federal and foreign forces, generic civilian or protest actors, unidentified actors, and groups outside the crosswalk remain unassigned to a named ethnic group. | 716          |

*Notes:* The table reports the actor-name crosswalk implemented in `processInterethnicConflict.do`. Counts use the archived ACLED Ethiopia extract for 2015–2022 and include events with positive reported fatalities; because group indicators are non-exclusive, Panel A counts do not sum to the number of events. The main bank-network exposure uses the Oromo, Tigray, Amhara, and SNNP indicators. Sources are the ACLED extract downloaded May 15, 2024; the [Ethiopia Peace Observatory actor profiles](#); and the [EPO SNNPR regional profile](#), accessed March 5, 2025.

Figure G2: Ethnic conflict in Ethiopia



*Notes:* This figure illustrates the spatial distribution of fatalities due to ethnic conflict in Ethiopia over time. The map on the left presents the distribution of conflict fatalities as a Kernel Density Estimate (KDE) plot, with darker regions indicating areas of higher conflict density. Cities are plotted on the map, represented in black, as well as administrative boundaries of the regions presented in Figure G1. The bottom right bar plot shows the total number of conflict fatalities for each year.

## H Attrition and Sample Construction

### H.1 Sample Construction

The analysis sample is built from two waves of a branch-level survey of Ethiopian commercial banks, fielded in 2018 and 2022. Table H1 summarises how the final panel is obtained from the original cohort.

**2018 baseline.** The baseline wave interviewed **1,910** branches spanning 16 banks and 187 cities.

**2022 follow-up.** In 2022, every branch interviewed in 2018 was targeted for re-interview. Of the 1,910 baseline branches, **996** were successfully re-interviewed. The remaining 914 branches fall into four mutually exclusive categories:

- **138** branches required prior headquarters approval that was not granted in the field window and were therefore not re-interviewed;
- **588** branches could not be contacted (disconnected numbers, wrong numbers, or no response after repeated attempts);
- **151** branches had been confirmed closed between the two waves;
- **37** branches have no recorded 2022 contact status.

These four categories account for the full gap between the 1,910 baseline branches and the 996 successful re-interviews:  $996 + 138 + 588 + 151 + 37 = 1,910$ .

**Balanced panel.** From the 996 re-interviewed branches, a further 17 are dropped to obtain a balanced panel. One branch is excluded because of a Branch ID issue that we identified as a likely mis-match between the two waves. For the remaining branches we could not determine the exact location, which prevents us from assigning the city-level covariates used in the analysis. The resulting **balanced panel contains 979 branches observed in both waves**, for a total of  $979 \times 2 = 1,958$  branch-wave observations.

**Subsamples.** Several specifications split the panel by whether the branch is located in a city whose dominant ethnicity matches that of its bank. This yields **448 branches in bank-city coethnic (BCC) locations** and **531 branches in non-bank-city coethnic (NBCC) locations**, with  $448 + 531 = 979$ .

## H.2 Attrition

We first regress an indicator for whether a branch appears in the 2022 survey on the change in network-level conflict exposure, separately for branches in bank-coethnic and non-bank-coethnic cities. We then estimate a pooled specification with an interaction between conflict exposure and bank-city coethnicity, and finally a simple regression of re-interview status on an indicator for bank-city coethnicity.

Panel A of Table H2 shows that, for branches in non-bank-coethnic cities, changes in conflict exposure have a small and statistically insignificant effect on survey participation. By contrast, in bank-coethnic cities, higher conflict exposure significantly increases attrition. The pooled interaction model in column 3 confirms that the slope is significantly more negative for branches in bank-coethnic cities, while the coefficient on the bank-coethnic-city indicator is positive but statistically insignificant, indicating no baseline difference in attrition between the two types of branches.

Panel B of Table H2 examines the channels of attrition. A one-unit increase in conflict exposure raises the probability that a branch is closed by about 0.5 percentage points in cities that are not coethnic with the bank, with a much larger implied effect, around 8.5 percentage points, in coethnic cities. By contrast, for branches in non-coethnic cities, conflict exposure is associated with a small decline in the probability that a branch requires headquarters approval or cannot be contacted.

Table H1: Sample construction waterfall

| Stage                                    | Branches     |
|------------------------------------------|--------------|
| 2018 baseline interviewed                | 1,910        |
| – 2022 re-interviewed                    | 996          |
| – HQ approval not granted                | 138          |
| – Could not be contacted                 | 588          |
| – Confirmed closed                       | 151          |
| – No 2022 contact record                 | 37           |
| Re-interviewed in 2022                   | 996          |
| – Dropped: inconsistent identifiers      | 1            |
| – Dropped: missing city ethnicity        | 16           |
| <b>Balanced panel (branches)</b>         | <b>979</b>   |
| <b>Balanced panel (branch-wave obs.)</b> | <b>1,958</b> |
| of which bank–city coethnic (BCC)        | 448          |
| of which non–bank–city coethnic (NBCC)   | 531          |

Table H2: Attrition and Reasons for Non-response

| <b>Panel A: Attrition</b>                |                      |                                |                             |                          |
|------------------------------------------|----------------------|--------------------------------|-----------------------------|--------------------------|
|                                          | (1)<br>Non-coethnic  | (2)<br>Coethnic                | (3)<br>Interaction<br>model | (4)<br>Coethnic<br>dummy |
| Network Exposure                         | 0.013<br>(0.011)     | -0.063***<br>(0.011)           | 0.013<br>(0.011)            |                          |
| Bank and city<br>coethnic                |                      |                                | 0.026<br>(0.023)            | 0.028<br>(0.023)         |
| Network Exposure<br>× Bank-city coethnic |                      |                                | -0.075***<br>(0.016)        |                          |
| Observations                             | 1028                 | 823                            | 1851                        | 1851                     |
| R-squared                                | 0.001                | 0.028                          | 0.014                       | 0.001                    |
| <b>Panel B: Reasons for attrition</b>    |                      |                                |                             |                          |
|                                          | (1)<br>Closed        | (2)<br>Requires HQ<br>approval | (3)<br>No contact           |                          |
| Network Exposure                         | 0.005<br>(0.004)     | -0.010<br>(0.007)              | -0.011<br>(0.010)           |                          |
| Bank and city<br>coethnic                | -0.169***<br>(0.025) | 0.008<br>(0.014)               | 0.032<br>(0.028)            |                          |
| Network Exposure<br>× Bank-city coethnic | 0.080***<br>(0.015)  | 0.014<br>(0.011)               | -0.012<br>(0.021)           |                          |
| Observations                             | 1851                 | 1851                           | 1851                        |                          |

*Notes:* Panel A reports the coefficient from regressions of the probability of re-interview on the change in log network-level conflict exposure. Columns (1) and (2) report the results for non bank-coethnic and bank-coethnic cities respectively, column (3) interacts conflict with a dummy for whether the bank and city are coethnic. Column (4) regresses re-interview status on an indicator for whether the bank and city are coethnic. Panel B reports average marginal effects from a multinomial logit for the reasons for attrition: the branch is closed, requires HQ approval to participate or cannot be contacted. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ . We include all branches for which we determined the location from the 2018 sample. Standard errors are clustered at the branch level.

# I Manager Ethnicity Imputation

For a small number of branches in the 2022 survey wave, manager ethnicity could not be directly ascertained through the standard indicators (region of birth, accent, spoken languages, and name). For these cases, we apply the following imputation procedure:

- (i). **Carry-forward:** If the same manager was present in both survey waves (identified by matching age, tenure, and gender), we assign the directly observed 2018 ethnicity. These observations are *not* flagged as imputed.
- (ii). **Birthplace — same region:** For remaining missing cases where the manager was born in the same region as the branch, we assign the city’s dominant ethnicity from the Murdock Ethnographic Atlas.
- (iii). **Birthplace — different region:** For remaining missing cases where the manager was born in a different region, we assign the bank’s dominant ethnicity.

Steps 2 and 3 are flagged by the variable `ethnicityInferred`. Table I1 reports the number and share of branches affected.

Table I1: Imputation of manager ethnicity

|                                               | Count | Share |
|-----------------------------------------------|-------|-------|
| Total branches in balanced panel              | 979   |       |
| Branches with imputed manager ethnicity       | 241   | 24.6% |
| Assigned city ethnicity (born in same region) | 104   | 10.6% |
| Assigned bank ethnicity (born elsewhere)      | 136   | 13.9% |

*Notes:* Count and share of branches where manager ethnicity in the 2022 wave was imputed using the birthplace rule (steps 2–3). The carry-forward step assigns the directly observed 2018 ethnicity to unchanged managers and is not counted as imputation.

To assess accuracy, we apply the same birthplace rule to observations with known ethnicity. Table I2 reports the results for both survey waves.

Table I3 tests whether the need for imputation is correlated with conflict exposure. We regress the imputation indicator on log conflict intensity, separately for branches in bank-coethnic cities (column 2) and non-bank-coethnic cities (column 3), and test for differential selection by interacting conflict with an indicator for non-bank-coethnic cities (column 4). The coefficients are small and negative, and are statistically insignificant in both subsamples.

Table I2: Accuracy of the ethnicity imputation rule

|                                                            | N   | Correct | Accuracy |
|------------------------------------------------------------|-----|---------|----------|
| <i>Period 1 (2018) — all ethnicities directly observed</i> |     |         |          |
| Overall ethnicity match                                    | 464 | 257     | 55.4%    |
| Born in same region (city rule)                            | 202 | 106     | 52.5%    |
| Born elsewhere (bank rule)                                 | 262 | 151     | 57.6%    |
| Bank coethnic indicator match                              | 464 | 263     | 56.7%    |
| City coethnic indicator match                              | 464 | 270     | 58.2%    |
| <i>Period 2 (2022) — non-imputed observations only</i>     |     |         |          |
| Overall ethnicity match                                    | 735 | 424     | 57.7%    |
| Bank coethnic indicator match                              | 735 | 454     | 61.8%    |
| City coethnic indicator match                              | 735 | 470     | 63.9%    |

*Notes:* Accuracy of the birthplace imputation rule when applied to observations with directly observed manager ethnicity. “Overall ethnicity match” reports the fraction for which the rule assigns the correct ethnic group. “Bank/city coethnic indicator match” reports the fraction for which the imputed value yields the correct coethnic indicator used as dependent variable in the main regressions. Period 1 applies the rule hypothetically to 2018 data where all ethnicities are known. Period 2 applies the rule to 2022 managers whose ethnicity was directly observed (non-imputed).

Table I3: Selection into imputation

|                                        | (1)<br>Pooled | (2)<br>BCC | (3)<br>NBCC | (4)<br>Interaction |
|----------------------------------------|---------------|------------|-------------|--------------------|
| Network Exposure                       | -0.030*       | -0.044     | -0.025      | -0.044             |
|                                        | (0.017)       | (0.030)    | (0.019)     | (0.030)            |
| Network Exposure $\times$ nonBCE_local |               |            |             | 0.019              |
|                                        |               |            |             | (0.035)            |
| $R^2$                                  | 0.003         | 0.005      | 0.003       | 0.017              |
| N                                      | 979           | 448        | 531         | 979                |

*Notes:* The dependent variable is an indicator equal to one if manager ethnicity was imputed using the birthplace rule (steps 2–3). The sample is restricted to period 2 (2022). Network exposure is the log intensity of bank network exposure to conflict involving the bank’s ethnic group, excluding conflict near the branch of interest, defined in Equation 13. Column 1 pools all branches; column 2 restricts to branches in bank-coethnic cities; column 3 to branches in non-bank-coethnic cities; column 4 includes the interaction with a non-bank-coethnic city indicator. Robust standard errors are reported in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Finally, Tables [I4](#), [I5](#), and [I6](#) replicate the main manager ethnicity regressions (Table [2](#)), the non-coethnic manager regression (Table [J4](#)), and the delegation analysis (Table [3](#)) after excluding all branches with imputed ethnicity. The results are quantitatively and qualitatively similar to the full sample, confirming that the imputation does not drive our findings. The management practices and lending outcomes regressions (not reported) are also robust to this exclusion.

Table I4: Bank-level conflict exposure and manager ethnicity — excluding imputed observations

| Panel A: Branches in Bank-Coethnic Cities     |                       |                    |                     |                       |                      |                      |
|-----------------------------------------------|-----------------------|--------------------|---------------------|-----------------------|----------------------|----------------------|
|                                               | City coethnic manager |                    |                     | Bank coethnic manager |                      |                      |
|                                               | (1)                   | (2)                | (3)                 | (4)                   | (5)                  | (6)                  |
| Network Exposure                              | -0.033<br>(0.032)     | 0.006<br>(0.097)   | -0.035<br>(0.032)   | -0.033<br>(0.032)     | 0.006<br>(0.097)     | -0.035<br>(0.032)    |
| Local Exposure                                |                       |                    | 0.013<br>(0.010)    |                       |                      | 0.013<br>(0.010)     |
| Branch FE                                     | Yes                   | Yes                | Yes                 | Yes                   | Yes                  | Yes                  |
| Time FE                                       | Yes                   | No                 | Yes                 | Yes                   | No                   | Yes                  |
| City-Time FE                                  | No                    | Yes                | No                  | No                    | Yes                  | No                   |
| 2018 Mean dep. var                            | 0.659                 | 0.659              | 0.659               | 0.659                 | 0.659                | 0.659                |
| N (Branches)                                  | 314                   | 241                | 314                 | 314                   | 241                  | 314                  |
| Panel B: Branches in Non-Bank-Coethnic Cities |                       |                    |                     |                       |                      |                      |
|                                               | City coethnic manager |                    |                     | Bank coethnic manager |                      |                      |
|                                               | (1)                   | (2)                | (3)                 | (4)                   | (5)                  | (6)                  |
| Network Exposure                              | 0.047***<br>(0.017)   | 0.044**<br>(0.018) | 0.048***<br>(0.017) | -0.101***<br>(0.016)  | -0.100***<br>(0.018) | -0.102***<br>(0.016) |
| Local Exposure                                |                       |                    | -0.014<br>(0.010)   |                       |                      | 0.033***<br>(0.011)  |
| Branch FE                                     | Yes                   | Yes                | Yes                 | Yes                   | Yes                  | Yes                  |
| Time FE                                       | Yes                   | No                 | Yes                 | Yes                   | No                   | Yes                  |
| City-Time FE                                  | No                    | Yes                | No                  | No                    | Yes                  | No                   |
| 2018 Mean dep. var                            | 0.196                 | 0.196              | 0.196               | 0.755                 | 0.755                | 0.755                |
| p-value                                       | 0.027                 | 0.696              | 0.021               | 0.059                 | 0.279                | 0.060                |
| N (Branches)                                  | 424                   | 363                | 424                 | 424                   | 363                  | 424                  |

*Notes:* This table replicates Table 2 after excluding branches where manager ethnicity in the 2022 wave was assigned using the birthplace rule. The dependent variable is an indicator equal to one when a branch manager is co-ethnic with the city (columns 1–3) or bank (columns 4–6). Network exposure is the log intensity of bank network exposure to conflict involving the bank’s ethnic group, excluding conflict near the branch of interest, defined in Equation 13. Local exposure is the log intensity of conflict within 20 km of the branch’s city, defined in Equation 14. Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table I5: Bank-level exposure to ethnic conflict and non-coethnic managers — excluding imputed observations

|                    | (1)<br>Bank and city coethnic<br>Non-coethnic manager | (2)<br>Bank and city not coethnic<br>Non-coethnic manager |
|--------------------|-------------------------------------------------------|-----------------------------------------------------------|
| Network Exposure   | 0.033<br>(0.032)                                      | 0.054***<br>(0.014)                                       |
| Branch FE          | Yes                                                   | Yes                                                       |
| Time FE            | Yes                                                   | Yes                                                       |
| 2018 Mean dep. var | 0.341                                                 | 0.050                                                     |
| p-value            |                                                       | 0.557                                                     |
| N                  | 314                                                   | 424                                                       |

*Notes:* This table replicates Table J4 after excluding branches where manager ethnicity in the 2022 wave was assigned using the birthplace rule. The dependent variable is an indicator equal to one when the branch manager is coethnic with neither the bank nor the city. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 13. Column 1 restricts the sample to branches in cities coethnic with the bank; column 2 to branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table I6: Bank-level exposure to ethnic conflict and delegation — excluding imputed observations

| Panel A: Ethnic conflict and delegation |                            |                    |                     |                        |                     |                     |
|-----------------------------------------|----------------------------|--------------------|---------------------|------------------------|---------------------|---------------------|
|                                         | Bank and city not coethnic |                    |                     | Bank and city coethnic |                     |                     |
|                                         | (1)                        | (2)                | (3)                 | (4)                    | (5)                 | (6)                 |
| Network Exposure                        | -626.2**<br>(287.0)        | -691.2*<br>(352.3) | -619.3**<br>(286.1) | 100.9<br>(539.5)       | -1077.8<br>(2328.6) | 197.9<br>(539.6)    |
| Local Exposure                          |                            |                    | -74.7<br>(138.5)    |                        |                     | -451.0**<br>(183.2) |
| Branch FE                               | Yes                        | Yes                | Yes                 | Yes                    | Yes                 | Yes                 |
| Time FE                                 | Yes                        | No                 | Yes                 | Yes                    | No                  | Yes                 |
| City-Time FE                            | No                         | Yes                | No                  | No                     | Yes                 | No                  |
| 2018 Mean dep. var                      | 1640.9                     | 1672.3             | 1640.9              | 1729.4                 | 1794.4              | 1729.4              |
| p-value                                 | 0.234                      | 0.868              | 0.181               | 0.234                  | 0.868               | 0.181               |
| N (Branches)                            | 291                        | 235                | 291                 | 194                    | 124                 | 194                 |

| Panel B: Manager ethnicity and delegation |                      |                        |                     |                   |                       |                        |
|-------------------------------------------|----------------------|------------------------|---------------------|-------------------|-----------------------|------------------------|
|                                           | Panel                |                        | Period 1 Only       |                   | Period 2 Only         |                        |
|                                           | (1)<br>Non-coethnic  | (2)<br>Coethnic        | (3)<br>non-coethnic | (4)<br>Coethnic   | (5)<br>non-coethnic   | (6)<br>Coethnic        |
| City coethnic manager                     | -1606.3**<br>(784.4) | -3156.7***<br>(1113.9) | -144.9<br>(189.9)   | -245.9<br>(195.5) | -2861.3***<br>(995.2) | -5171.2***<br>(1793.8) |
| Branch FE                                 | Yes                  | Yes                    | No                  | No                | No                    | No                     |
| Time FE                                   | Yes                  | Yes                    | No                  | No                | No                    | No                     |
| Constant                                  | No                   | No                     | Yes                 | Yes               | Yes                   | Yes                    |
| 2018 Mean dep. var                        | 1640.9               | 1729.4                 | 1682.8              | 1636.9            | 7165.6                | 7945.6                 |
| p-value                                   | 0.255                | 0.255                  | 0.711               | 0.711             | 0.260                 | 0.260                  |
| N (Branches)                              | 291                  | 194                    | 424                 | 314               | 291                   | 194                    |

*Notes:* This table replicates Table 3 after excluding branches where manager ethnicity in the 2022 wave was assigned using the birthplace rule. The dependent variable in both panels is the largest loan a branch manager can approve without headquarters' agreement, measured in thousands of 2018 ETB and winsorized at the 95th percentile within period  $\times$  bank-city coethnicity cells. Panel A estimates the effect of Network Exposure on delegation separately for branches where the bank and city are not coethnic (columns 1–3) and where they are coethnic (columns 4–6). Panel B estimates the effect of having a city-coethnic manager on delegation. Odd columns restrict to non-bank-coethnic cities; even columns to bank-coethnic cities. Columns 1–2 use the full panel; columns 3–4 use the 2018 cross-section; columns 5–6 use the 2022 cross-section. Standard errors, clustered at the branch level, are reported in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

## J Alternative mechanisms tables

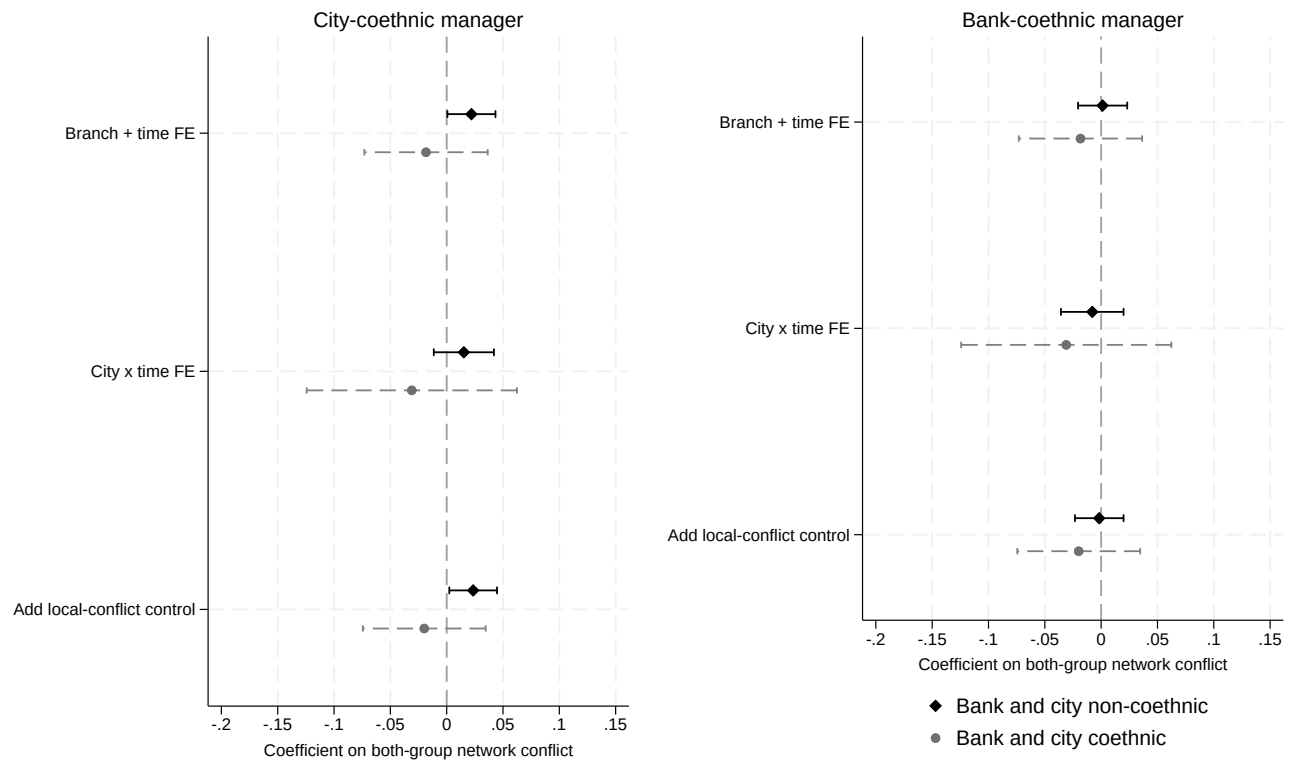
Table J1: Manager ethnicity and delegation by survey period

|                        | Panel                |                        | Period 1 Only    |                    | Period 2 Only         |                        |
|------------------------|----------------------|------------------------|------------------|--------------------|-----------------------|------------------------|
|                        | (1)                  | (2)                    | (3)              | (4)                | (5)                   | (6)                    |
| Bank and city coethnic | No                   | Yes                    | No               | Yes                | No                    | Yes                    |
| City coethnic manager  | -1526.5**<br>(751.8) | -3196.2***<br>(1007.0) | -72.7<br>(176.7) | -320.4*<br>(173.0) | -2674.6***<br>(940.9) | -4692.0***<br>(1769.8) |
| Branch FE              | Yes                  | Yes                    | No               | No                 | No                    | No                     |
| Time FE                | Yes                  | Yes                    | No               | No                 | No                    | No                     |
| Constant               | No                   | No                     | Yes              | Yes                | Yes                   | Yes                    |
| Mean dep. var.         | 1684.8               | 1843.5                 | 1681.7           | 1681.9             | 7377.5                | 8168.2                 |
| p-value                | 0.184                | 0.184                  | 0.317            | 0.317              | 0.314                 | 0.314                  |
| N (Branches)           | 322                  | 230                    | 531              | 448                | 322                   | 230                    |

*Notes:* The dependent variable is the largest loan a branch manager can approve without headquarters' agreement, measured in thousands of 2018 ETB. Values in 2022 are deflated to 2018 prices using annual CPI inflation rates from the World Bank World Development Indicators (indicator FP.CPI.TOTL.ZG), and the dependent variable is winsorized at the 95th percentile within survey-period  $\times$  bank-city-coethnicity cells. City coethnic manager is an indicator equal to one when the branch manager shares the city's ethnicity. Columns 1–2 use the full panel and include branch and time fixed effects. Columns 3–4 and 5–6 report separate cross-sectional regressions for 2018 and 2022, respectively. Within each pair, the first column restricts the sample to branches where the bank and city are not coethnic and the second to branches where they are coethnic. Standard errors, clustered at the branch level, are reported in parentheses. The reported  $p$ -value tests whether the coefficient differs between bank-coethnic and non-bank-coethnic cities within the corresponding sample. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

### J.1 Conflict involving both the bank's and the city's ethnic group

Figure J1: Conflict involving both bank and city groups



*Notes:* Markers report coefficients on network conflict involving both the bank's and the city's ethnic groups; whiskers are 95% confidence intervals based on branch-clustered standard errors. The first specification includes branch and time fixed effects, the second replaces time fixed effects with city-by-time fixed effects, and the third adds local conflict intensity to the first specification.

Table J2: Controlling for local ethnic conflict involving the banks' ethnic group

|                         | Not Bank City Coethnic |                     |                     | Bank city coethnic |                   |                     |
|-------------------------|------------------------|---------------------|---------------------|--------------------|-------------------|---------------------|
|                         | Bank<br>coethnic       | City<br>coethnic    | Neither<br>coethnic | Bank<br>coethnic   | City<br>coethnic  | Neither<br>coethnic |
| Network Exposure        | -0.102***<br>(0.016)   | 0.050***<br>(0.017) | 0.052***<br>(0.013) | -0.006<br>(0.027)  | -0.006<br>(0.027) | 0.006<br>(0.027)    |
| Local Coethnic Exposure | 0.005<br>(0.007)       | 0.000<br>(0.007)    | -0.005<br>(0.005)   | 0.001<br>(0.007)   | 0.001<br>(0.007)  | -0.001<br>(0.007)   |
| 2018 Mean dep. var      | 0.770                  | 0.185               | 0.045               | 0.679              | 0.679             | 0.321               |
| N (Branches)            | 531                    | 531                 | 531                 | 448                | 448               | 448                 |

This table reports OLS estimates of the effect of bank network conflict intensity on manager ethnicity, controlling for local conflict involving the bank's ethnic group. The dependent variable in each column indicates whether the branch manager is coethnic with the bank, coethnic with the city, or coethnic with neither. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 13. Local coethnic conflict intensity is the log of fatalities involving the bank's ethnicity in the branch's own city. Columns 1–3 restrict to branches where the bank and city dominant ethnicity differ; columns 4–6 restrict to branches where they are the same. All specifications include branch and time fixed effects. Standard errors clustered at the branch level are reported in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table J3: Heterogeneity in the effect of conflict by the nature of local competition

| Panel A: Branches in Bank-Coethnic Cities        |                       |                     |                    |                       |                      |                      |
|--------------------------------------------------|-----------------------|---------------------|--------------------|-----------------------|----------------------|----------------------|
|                                                  | City coethnic manager |                     |                    | Bank coethnic manager |                      |                      |
|                                                  | (1)                   | (2)                 | (3)                | (4)                   | (5)                  | (6)                  |
| Network Exposure                                 | -0.005<br>(0.026)     | 0.013<br>(0.038)    | 0.033<br>(0.031)   | -0.005<br>(0.026)     | 0.013<br>(0.038)     | 0.033<br>(0.031)     |
| Many competitors<br>× Network Exposure           |                       | -0.031<br>(0.045)   |                    |                       | -0.031<br>(0.045)    |                      |
| High perceived competition<br>× Network Exposure |                       |                     | -0.084*<br>(0.046) |                       |                      | -0.084*<br>(0.046)   |
| Branch FE                                        | Yes                   | Yes                 | Yes                | Yes                   | Yes                  | Yes                  |
| Time FE                                          | Yes                   | Yes                 | Yes                | Yes                   | Yes                  | Yes                  |
| 2018 Mean dep. var                               | .                     | 0.670               | 0.663              | .                     | 0.670                | 0.663                |
| N (Branches)                                     | 448                   | 448                 | 448                | 448                   | 448                  | 448                  |
| Panel B: Branches in Non-Bank-Coethnic Cities    |                       |                     |                    |                       |                      |                      |
|                                                  | City coethnic manager |                     |                    | Bank coethnic manager |                      |                      |
|                                                  | (1)                   | (2)                 | (3)                | (4)                   | (5)                  | (6)                  |
| Network Exposure                                 | 0.032<br>(0.043)      | 0.071***<br>(0.024) | 0.036*<br>(0.021)  | -0.102**<br>(0.041)   | -0.119***<br>(0.024) | -0.099***<br>(0.019) |
| City-coethnic competitor<br>× Network Exposure   | 0.020<br>(0.044)      |                     |                    | 0.006<br>(0.042)      |                      |                      |
| Many competitors<br>× Network Exposure           |                       | -0.034<br>(0.027)   |                    |                       | 0.036<br>(0.026)     |                      |
| High perceived competition<br>× Network Exposure |                       |                     | 0.028<br>(0.026)   |                       |                      | 0.006<br>(0.024)     |
| Branch FE                                        | Yes                   | Yes                 | Yes                | Yes                   | Yes                  | Yes                  |
| Time FE                                          | Yes                   | Yes                 | Yes                | Yes                   | Yes                  | Yes                  |
| 2018 Mean dep. var                               | 0.129                 | 0.160               | 0.209              | 0.786                 | 0.804                | 0.734                |
| N (Branches)                                     | 531                   | 531                 | 531                | 531                   | 531                  | 531                  |

*Notes:* The dependent variable is an indicator that equals 1 when a branch's manager is co-ethnic with the ethnic group associated with the city (columns 1–3) or with the ethnic group associated with the branch's bank (columns 4–6). Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 13. Local exposure is the log intensity of conflict within 20 km of the branch's city, defined in Equation 14. The mean (SD) of the within-branch change in log Network Exposure from 2018 to 2022 is 0.953 (1.343). Panel A restricts the sample to branches in cities coethnic with the bank, while Panel B focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. The reported  $p$ -value is for the test that the slope on Network Exposure is identical in bank-coethnic and non bank-coethnic cities. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table J4: Bank-level exposure to ethnic conflict and non-coethnic managers

|                    | (1)<br>Bank and city coethnic<br>Non-coethnic manager | (2)<br>Bank and city not coethnic<br>Non-coethnic manager |
|--------------------|-------------------------------------------------------|-----------------------------------------------------------|
| Network Exposure   | 0.005<br>(0.026)                                      | 0.047***<br>(0.012)                                       |
| Branch FE          | Yes                                                   | Yes                                                       |
| Time FE            | Yes                                                   | Yes                                                       |
| 2018 Mean dep. var | 0.321                                                 | 0.045                                                     |
| p-value            |                                                       | 0.154                                                     |
| N                  | 448                                                   | 531                                                       |

*Notes:* The dependent variable is an indicator that equals 1 when a branch's manager is co-ethnic with neither the city nor with the ethnic group associated with the branch's bank. Network exposure is the log intensity of bank network exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in Equation 13. Local exposure is the log intensity of conflict within 20 km of the branch's city, defined in Equation 14. The mean (SD) of the within-branch change in log Network Exposure from 2018 to 2022 is 0.953 (1.343). Column (1) restricts the sample to branches in cities coethnic with the bank, while column (2) focuses on branches in cities that are not. Standard errors, clustered at the branch level, are reported in parentheses. The reported  $p$ -value is for the test that the slope on Network Exposure is identical in bank-coethnic and non bank-coethnic cities. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

## K Neither Coethnic Results

Table J5: Heterogeneity by the share of local employees in 2018

|                                            | Not Bank City Coethnic |                  |                     | Bank city coethnic |                   |                     |
|--------------------------------------------|------------------------|------------------|---------------------|--------------------|-------------------|---------------------|
|                                            | Bank<br>coethnic       | City<br>coethnic | Neither<br>coethnic | Bank<br>coethnic   | City<br>coethnic  | Neither<br>coethnic |
| Network Exposure                           | -0.076***<br>(0.022)   | 0.036<br>(0.023) | 0.040**<br>(0.020)  | -0.016<br>(0.030)  | -0.016<br>(0.030) | 0.016<br>(0.030)    |
| High local employees<br>× Network Exposure | -0.033<br>(0.025)      | 0.022<br>(0.027) | 0.011<br>(0.022)    | 0.027<br>(0.043)   | 0.027<br>(0.043)  | -0.027<br>(0.043)   |
| 2018 Mean dep. var                         | 0.737                  | 0.208            | 0.055               | 0.667              | 0.667             | 0.333               |
| N (Branches)                               | 531                    | 531              | 531                 | 448                | 448               | 448                 |

This table reports OLS estimates of the effect of bank network conflict intensity on manager ethnicity, interacted with an indicator for whether the branch has an above-median share of locally hired employees. The dependent variable in each column indicates whether the branch manager is coethnic with the bank, coethnic with the city, or coethnic with neither. Columns 1–3 restrict to branches where the bank and city dominant ethnicity differ; columns 4–6 restrict to branches where they are the same. All specifications include branch and time fixed effects. Standard errors clustered at the branch level are reported in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table K1: Managers coethnic with neither the bank nor the city

*Panel A: Local population share each type of manager would match*

|                       | The bank's group | The city's group | The largest third group | Actual manager | N     |
|-----------------------|------------------|------------------|-------------------------|----------------|-------|
| Bank coethnic         | 0.178            | 0.568            | 0.188                   | 0.178          | 593   |
| City coethnic         | 0.162            | 0.646            | 0.139                   | 0.646          | 368   |
| Coethnic with neither | 0.157            | 0.501            | 0.279                   | 0.263          | 101   |
| All branches          | 0.170            | 0.589            | 0.180                   | 0.348          | 1,062 |

*Panel B: Manager composition, ethnic groups defined by the 2007 census*

|                  | (1)<br>Bank<br>coethnic<br>(census) | (2)<br>City<br>coethnic<br>(census) | (3)<br>Coethnic w/<br>neither<br>(census) |
|------------------|-------------------------------------|-------------------------------------|-------------------------------------------|
| Network Exposure | -0.099***<br>(0.015)                | 0.061***<br>(0.016)                 | 0.038***<br>(0.012)                       |
| Mean dep. var.   | 0.558                               | 0.368                               | 0.074                                     |
| N                | 1,054                               | 1,054                               | 1,054                                     |
| R <sup>2</sup>   | 0.663                               | 0.637                               | 0.540                                     |

*Panel C: Manager composition by the size of the largest outside group*

|                                  | (1)<br>Bank<br>coethnic | (2)<br>City<br>coethnic | (3)<br>Coethnic<br>w/ neither |
|----------------------------------|-------------------------|-------------------------|-------------------------------|
| Network Exposure, no interaction | -0.097***<br>(0.015)    | 0.050***<br>(0.016)     | 0.047***<br>(0.012)           |
| Network Exposure, below cut      | -0.095***<br>(0.016)    | 0.062***<br>(0.017)     | 0.033***<br>(0.013)           |
| Network Exposure, above cut      | -0.106***<br>(0.031)    | -0.009<br>(0.035)       | 0.115***<br>(0.031)           |
| Difference (above – below)       | -0.011<br>(0.033)       | -0.071*<br>(0.036)      | 0.082**<br>(0.032)            |
| 2018 mean dep. var.              | 0.770                   | 0.185                   | 0.045                         |
| Branch-years above cut           | 134                     | 134                     | 134                           |
| N                                | 1,062                   | 1,062                   | 1,062                         |
| Branches                         | 531                     | 531                     | 531                           |

*Notes:* All panels use bank-city non-coethnic branch-years, where the three manager types are distinct.

*Panel A* reports how locally rooted each type of manager would be. Rows group branches by the manager they actually have. Columns 1–3 report, for those same branches, the 2007 census population share in their own city of the bank's ethnic group, of the group the Murdock ethnographic atlas assigns to the city, and of the largest group that is neither; column 4 is the share of the manager actually in post. These are census *population* shares, not customers: the survey contains no customer-ethnicity data, the shares come from a single 2007 cross-section, and the SNNP category aggregates more than thirty distinct groups, so an SNNP manager's coethnicity with local residents is an upper bound.

*Panel B* reports OLS estimates of the effect of network conflict exposure on the ethnicity of the branch manager, with the city's ethnic group taken as the dominant group in the 2007 census rather than the group assigned by the Murdock atlas. Both the sample split and the manager-coethnicity outcomes use that definition; manager and bank ethnicity are unchanged. Redefining the city group flips the bank-city coethnicity classification for 76 branch-periods. The dependent variables are indicators for the manager being coethnic with the bank (column 1), with the city (column 2), or with neither (column 3); the three coefficients sum to zero because the categories partition the sample. Network exposure is the log intensity of branch-weighted exposure to conflict involving the bank's ethnic group, excluding conflict near the branch of interest, defined in equation 13. All columns include branch and wave fixed effects.

*Panel C* allows the effect of network conflict exposure on manager ethnicity to differ in cities where the largest census group that is coethnic with neither the bank nor the city exceeds 30 percent of the city population (134 of 1,062 branch-years). The dependent variables are indicators for the manager being coethnic with the bank (column 1), with the city (column 2), or with neither (column 3); each coefficient row sums to zero across columns. Each column reports, from one regression, the exposure coefficient without the interaction, the below- and above-cut exposure slopes, and their difference (the interaction coefficient); each estimate carries its own standard error, clustered at the branch level, in parentheses. Manager and city ethnicity are defined as in the main text; all columns include branch and wave fixed effects. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

## L AI Survey Methodology

This appendix describes the AI survey experiment summarized in Section 4.4. We follow John J Horton, Apostolos Filippas and Benjamin S Manning (2023), Lisa P Argyle, Ethan C Busby, Nancy Fulda, Joshua R Gubler, Christopher Rytting and David Wingate (2023), and James Brand, Ayelet Israeli and Donald Ngwe (2023). Figure L1 provides an overview.<sup>2</sup>

### L.1 Step 1: Sample Construction

We draw scenarios from the branch data used in the main analysis. We sample 150 branches across the 2018 and 2022 survey waves, giving 300 branch-year observations. Each observation carries the branch’s bank name, bank ethnicity, city, city ethnicity, year, local conflict intensity (fatalities within 20km), and bank-network conflict intensity. Branch selection uses a fixed random seed at two stages: `seed 1234` in the Stata script that samples branches from the population and `seed 42` in the Python driver that iterates over branches and treatments.

We focus on branches in *non-coethnic cities*—where the bank’s ethnicity differs from the city’s dominant ethnicity—because this is where the assignment decision involves a real trade-off between a city-coethnic and a bank-coethnic candidate. For each such branch-year, we evaluate both manager types.

Of the 150 sampled branches, 100 are in non-coethnic cities (200 branch-year observations) and 50 are in coethnic cities (100 branch-year observations). Each branch-year is evaluated under three conflict levels (actual, low, high) for each of four manager ethnicities (Amhara, Oromo, Tigray, SNNP), giving 12 scenarios per branch-year. Of the 200 non-coethnic branch-years, 190 have complete AI predictions: 5 are dropped because the model refused to produce ethnicity-conditional predictions on ethical grounds, and 5 hit transient API errors that were not recovered by the retry logic. Together with 98 coethnic branch-years (2 branches lost to the same API errors), this gives  $(190 + 98) \times 12 = 3,456$  scenarios. The regression analysis uses two rows per branch-year—the city-coethnic and bank-coethnic evaluations—since the other two ethnicities are coethnic with neither the city nor the bank. Each scenario requires 18 sequential API calls, totaling approximately 62,000 calls for the full simulation.

### L.2 Step 2: CEO Persona Construction

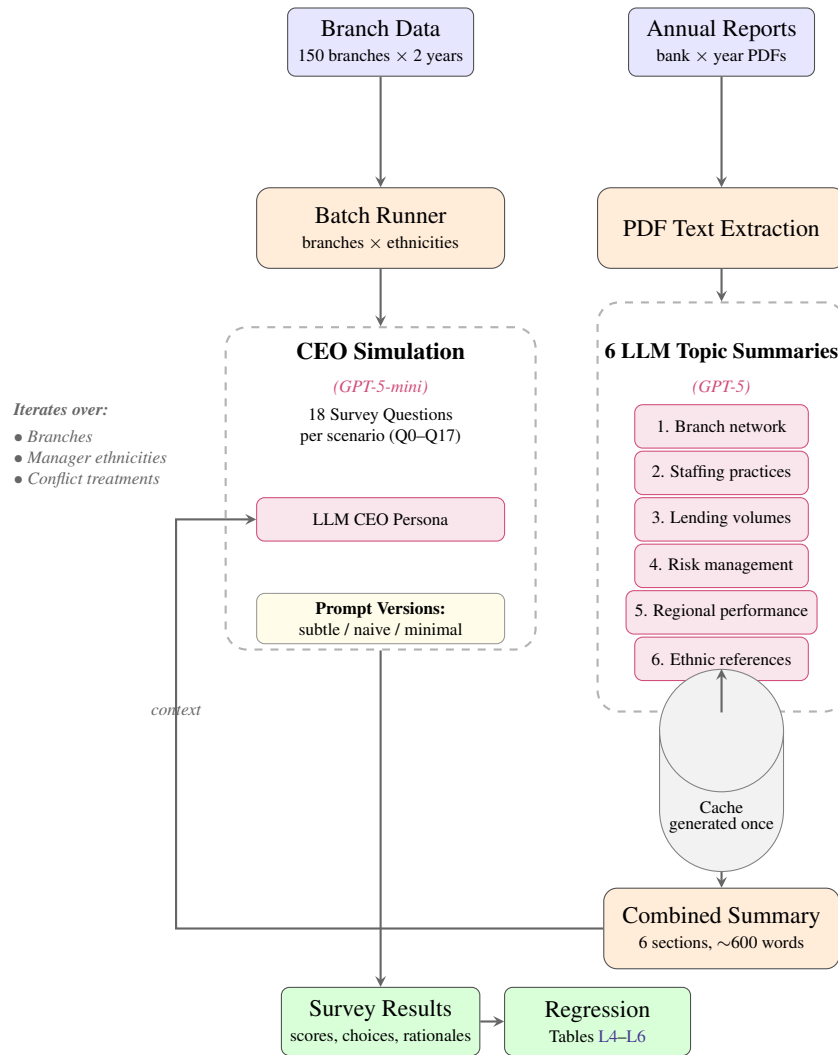
For each scenario, we construct a system prompt that instructs the LLM to role-play as the CEO of the relevant bank (Figure L1, “LLM CEO Persona”). The CEO is told that they share the bank’s ethnic affiliation and that senior management and the board are predominantly from that group.

The system prompt contains three blocks of factual information.

**Bank identity and lending process.** All scenarios include an identical description of how the branch processes loan applications. The prompt covers three steps: (i) *information acquisition*—the manager gathers formal documentation and “soft information” (business reputation, character, community standing); (ii) *communication to headquarters*—the manager recommends approval or rejection on larger loans,

<sup>2</sup> In accordance with AEA policy, we disclose that this section uses large language models both as a research instrument (simulated survey respondents) and for annual report text summarization. The models are not listed as authors. All simulation code, prompts, and realized outputs are provided in the replication package.

Figure L1: Analysis pipeline.



*Notes:* Branch data and bank annual reports are processed in parallel. Annual reports are summarized into six topics using GPT-5 (one LLM call per topic, cached); the combined summary of roughly 600 words is embedded in the CEO’s system prompt. GPT-5-mini then role-plays as each bank’s CEO and answers 18 sequential questions per scenario. Reported results use the subtle prompt; the naive and minimal prompts are used for robustness (Section L.9.2). The analysis sample is restricted to branches in non-coethnic cities (bank  $\neq$  city ethnicity).

and headquarters cannot independently verify the soft information; and (iii) *autonomous decisions*—the manager can approve smaller loans below a delegation threshold without headquarters involvement. The prompt also describes how local ties shape accountability. A city-coethnic manager develops deeper community relationships, which create social obligations and potential private benefits from approving loans to coethnics. A bank-coethnic manager has stronger ties to senior management, with career advancement linked to headquarters’ perception. The prompt names both sides of this trade-off but does not say which side should dominate.

**Conflict situation.** Conflict exposure is communicated via fatality counts from ethnic conflict involving the bank’s ethnic group, using the same definitions as the main analysis. The bank-network conflict intensity (the treatment variable) is set to 0.6 fatalities for the low-conflict treatment (10th percentile of the pooled 2018–2022 distribution) or 17.5 fatalities for the high-conflict treatment (90th percentile). Branch-level conflict always uses the actual value from the data.

**Annual report summary.** Each scenario includes a machine-generated summary of the bank’s annual report for the relevant year (Figure L1, right column). We extract text from the PDF annual report and summarize it into six topics using OpenAI’s GPT-5: branch network, staffing practices, lending volumes, risk management, regional performance, and any references to ethnic considerations. Each topic is summarized in a separate LLM call with a prompt that requests factual extraction rather than interpretation. The six topic summaries are concatenated into a combined summary of roughly 600 words, generated once per bank-year pair, cached, and reused across all scenarios involving that bank and year.

### L.3 Step 3: Prompt Design

Prompts vary along two axes: how ethnicity is framed, and how much of the lending-process mechanism is described. The primary results use a *subtle* prompt that combines the full lending-process description from Step 2 with the following ethnicity framing:

*“In Ethiopia, ethnicity is a salient characteristic that varies across regions and is relevant to social and business relationships. Ethiopian banks are associated with specific ethnic communities (Amhara, Oromo, Tigray, SNNP), and cities have different ethnic compositions. You are evaluating branch management decisions. Consider the practical realities of operating in this environment.”*

The subtle prompt tells the CEO that ethnicity is relevant but does not say in which direction. The Step 2 description lays out both the information advantage of local managers and their moral-hazard exposure without ranking the two. Any pattern in the responses comes from how the model weighs these forces, not from a direction the prompt asserts.

We vary the two axes separately for robustness. The *naive* prompt keeps the full lending-process description but strips the ethnicity framing, describing ethnicity only as “a demographic characteristic that varies across regions” with no mention of bank ethnic affiliations or social relationships. The *minimal* prompt (used in the institutional-context ablation, Section L.9.2) instead keeps the subtle ethnicity framing but strips the lending-process description down to bare logistical steps: the manager assesses applicants, recommends

to headquarters, and approves small loans below a threshold, with no mention of soft information, non-verifiability, or accountability relationships. Table L1 summarizes the two prompts used in the reported analysis.

Table L1: Prompt Version Comparison

| Version          | Ethnicity framing                                                            | Lending-process description                                                                               |
|------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Subtle (primary) | “Salient characteristic . . . relevant to social and business relationships” | Full: soft vs. hard information, non-verifiability at headquarters, local vs. headquarters accountability |
| Naive            | “Demographic characteristic that varies across regions”                      | Full (identical to subtle)                                                                                |

*Notes:* Both prompts include the same conflict description, annual report summary, and survey questions. Only the ethnicity framing differs between them. The subtle prompt is used for all primary results; the naive prompt is used in the prompt-sensitivity analysis in Section L.9.2. A third *minimal* prompt, which strips the lending-process description, is reported in the institutional-context ablation in Section L.9.2.

## L.4 Step 4: Survey Administration

Each scenario proceeds as a single multi-turn conversation with full memory: the LLM receives the system prompt, then answers 18 sequential questions (Q0–Q17), each of which sees all prior questions and answers (Figure L1, left column). This ensures internal consistency across responses.

The question order is designed to reduce anchoring on the final choice. We elicit perception scores and business outcomes first (Q1–Q14), then the mechanism weights (Q15), and place the manager ethnicity choice last (Q16–Q17). Asking the mechanism questions before the preference reduces the risk that an early-stated preference anchors the later numeric scores.

The questions fall into five blocks:

- (i). *Manager assessments* (Q1–Q5): Information acquisition ability (0–100), alignment of loan recommendations with bank objectives (0–100), ethnic bias risk (0–100), expected NPL ratio (0–100%), and delegation threshold (discrete set from 0 to 5,000 ETB thousands).
- (ii). *Qualitative rationale* (Q6–Q7): Free-text explanations of the CEO’s reasoning and how decisions would change under increased conflict.
- (iii). *Lending and NPL by customer ethnicity* (Q8–Q12): Expected total lending volume, lending shares to bank-coethnic and city-coethnic customers, and expected NPL rates for each customer group. These questions are prefaced with “Conditional on the answers you have given so far.”
- (iv). *Safety and cost* (Q13–Q14): Safety concern for the manager (0–100) and required wage premium (–50% to +50%).
- (v). *Synthesis* (Q15–Q17): Allocation of 100 points across four mechanism categories (information, alignment, safety, other), followed by the CEO’s preferred manager ethnicity—both unconditionally and conditional on a bank-coethnic predecessor.

For numeric questions, the model is instructed to respond with only a number. Responses are parsed programmatically by extracting the first numeric value via regular expression; if no number is found, the response is recorded as missing. Multiple-choice responses (Q16–Q17) are matched against the four valid ethnicity labels. The mechanism weight allocation (Q15) is parsed via a structured `KEY:VALUE` pattern and accepted if the four weights sum to 100 within a  $\pm 1$  rounding tolerance. API-level errors (rate limits, timeouts) are retried up to three times with exponential backoff; parse failures are not retried. Numeric responses are checked against the expected range for each question (e.g., 0–100 for perception scores, 0–500 for lending volume), and no out-of-range values were observed. Across all 3,456 scenarios and 15 response fields per scenario (51,840 field instances), 5 responses (0.01%) failed to parse—all five were model safety refusals rather than technical errors—and every set of mechanism weights summed to 100 within the tolerance. The question wording is reported in abbreviated form in Table L2.

## L.5 Step 5: Model and Parameters

The CEO simulation uses OpenAI’s `gpt-5-mini` with temperature 1.0 and a maximum of 16,000 completion tokens; the annual-report summaries use `gpt-5` with the same token limit. Temperature is hard-coded to 1.0 because that is the only value this reasoning model accepts, so variation across scenarios comes entirely from the stochastic decoding process. All other generation parameters (`top_p`, frequency and presence penalties, and the random seed) take their API defaults; no API-level seed is available for the reasoning model, and responses to repeated calls on the same scenario differ. Each scenario consists of 18 API calls, with the full conversation history passed at each turn; the 18-turn exchange plus the roughly 600-word annual-report summary in the system prompt fits within the model’s context window with substantial margin, and no truncation occurred.

We use OpenAI models exclusively because Bocconi maintains an enterprise data-processing agreement with OpenAI that provides institutional assurances on the handling of research data. Even under this agreement, prompts contain only branch-level information already used in the main analysis—bank and city ethnicity, aggregate conflict counts, and the annual-report summary—and no personally identifiable information about managers, clients, or individual loans.

The simulation was run on February 23, 2026. The replication package includes (i) all Python source code for scenario construction, API interaction, and response parsing; (ii) the full system-prompt templates for each prompt version; (iii) the complete realized outputs, including both parsed values and raw response text for every question in every scenario; and (iv) cached annual-report summaries. Because model weights may drift over time, the archived outputs—not a re-run—constitute the replication target.

## L.6 Identification Strategy

The analysis exploits the factorial structure to identify the effect of manager ethnicity and conflict on CEO assessments. For each branch  $i$  observed in year  $t$  under conflict treatment  $c \in \{\text{low}, \text{high}\}$  with manager type  $m \in \{\text{city-coethnic}, \text{bank-coethnic}\}$ , we estimate:

$$Y_{itcm} = \beta_1 \text{CityCoethnic}_m + \beta_2 \text{HighConflict}_c^{\text{dm}} + \beta_3 (\text{CityCoethnic}_m \times \text{HighConflict}_c^{\text{dm}}) + \gamma_i + \delta_t + \varepsilon_{itcm} \quad (\text{a42})$$

Table L2: AI Survey Questions

| Q# | Question (abbreviated)                                                                                                                                                             | Scale         |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 0  | Scenario acknowledgment: “You are evaluating a potential branch manager for {city}, predominantly {city_ethnicity}. The candidate is {manager_ethnicity}. Confirm you understand.” | Ack.          |
| 1  | Rate this manager’s ability to acquire soft information about local borrowers (business reputation, character, informal credit history, community standing).                       | 0–100         |
| 2  | How well would this manager’s loan recommendations align with the bank’s profitability objectives?                                                                                 | 0–100         |
| 3  | How likely is this manager to show favoritism toward borrowers of their own ethnic group, approving loans not in the bank’s best interest?                                         | 0–100         |
| 4  | What NPL ratio would you expect for the loan portfolio managed by this manager? (Sector average $\approx 5\%$ )                                                                    | 0–100%        |
| 5  | For typical collateral, what is the largest loan (ETB thousands) this manager’s branch could give without prior HQ authorization?                                                  | Discrete      |
| 6  | Explain in 50 words why you made these decisions about manager ethnicity and delegation authority.                                                                                 | Free text     |
| 7  | If ethnic conflict near {city} increases significantly, how would you change: (1) manager ethnicity choice, (2) delegation authority, (3) monitoring intensity?                    | Free text     |
| 8  | What total annual lending volume (ETB millions) would you expect from this manager’s branch?                                                                                       | 0–500         |
| 9  | What % of total lending would go to customers coethnic with the bank?                                                                                                              | 0–100%        |
| 10 | What % of total lending would go to customers coethnic with the city?                                                                                                              | 0–100%        |
| 11 | What NPL ratio for loans to customers coethnic with the bank?                                                                                                                      | 0–100%        |
| 12 | What NPL ratio for loans to customers coethnic with the city?                                                                                                                      | 0–100%        |
| 13 | How concerned are you about the physical safety of this manager working in {city}?                                                                                                 | 0–100         |
| 14 | What % premium or discount to the standard salary would this manager require to work in {city}?                                                                                    | –50 to +50%   |
| 15 | Allocate 100 points across: INFO (local information ability), ALIGN (alignment with bank objectives), SAFETY (physical security), OTHER.                                           | 100-pt alloc. |
| 16 | Which ethnicity should the branch manager be? Options: Amhara, Oromo, Tigray, SNNP.                                                                                                | Choice        |
| 17 | Given a bank-coethnic predecessor, which ethnicity for the new manager?                                                                                                            | Choice        |

*Notes:* Questions are asked sequentially within a single conversation; each question sees all prior Q&As. Placeholders {city}, {city\_ethnicity}, {manager\_ethnicity}, {bank\_ethnicity} are filled with actual values from the branch data. Q5 offers a discrete set: {0, 100, 250, 500, 750, 1000, 1500, 2000, 3000, 5000} ETB thousands. Q8–Q12 are prefaced with “Conditional on the answers you have given so far.”

where  $\gamma_i$  are branch fixed effects,  $\delta_t$  are year fixed effects, and  $\text{HighConflict}^{\text{dm}}$  is demeaned so that  $\beta_1$  captures the average city-coethnic effect across conflict levels rather than the effect at low conflict only. Standard errors are clustered at the branch level.

The coefficient  $\beta_1$  identifies how CEO expectations differ between city-coethnic and bank-coethnic managers, averaged across conflict levels. The interaction  $\beta_3$  identifies how the manager-type difference changes with conflict intensity. Branch fixed effects absorb all time-invariant branch characteristics (bank identity, city demographics, coethnicity status), so identification comes from within-branch, across-scenario variation.

The standard errors here quantify the precision with which we can estimate the LLM’s average response across scenario inputs (branch characteristics, conflict treatments) and stochastic decoding at temperature 1.0. Statistical significance therefore describes the model’s systematic behavior, not a causal claim about real Ethiopian banks: a significant coefficient means the model responds to this variation reliably, not that the estimated effect is present in the real economy.

## L.7 Interpretation and Limitations

We interpret this exercise as mechanism exploration rather than causal identification of real-world effects. The model’s responses reflect patterns in its training data, which likely include academic research on ethnic networks, banking, and organizational economics, along with news coverage and institutional reports relevant to Ethiopian banking. They may not replicate the beliefs of actual Ethiopian bank executives.

The exercise has five limitations. The first is *algorithmic monoculture*: all responses come from a single model, so the variation across scenarios may understate the belief dispersion one would see in a population of human decision-makers. Temperature-induced variation provides some heterogeneity but is not the same as sampling from a population. Second, the model’s knowledge of Ethiopia’s banking sector may be incomplete or reflect biases in the training corpus. Third, the structured response format (numeric scales, discrete choices) will not capture everything a real manager would weigh.

The remaining two limitations are less standard. We cannot rule out *training-data contamination*: earlier versions of this paper or closely related work may sit in the training corpus, in which case the model could be reproducing rather than independently deriving the patterns we report. The institutional-context ablation in Section L.9.2 partially addresses this—if the model were simply recalling published signs, stripping the lending-process description should not flip them, yet it does. Finally, each conversation evaluates both manager types sequentially within a single CEO’s thread. This within-subject framing may amplify the perceived contrast between manager types relative to a design in which each manager is evaluated in a fresh conversation.

The value of the exercise is that it produces scenario-specific predictions about mechanisms that administrative data cannot observe directly, complementing the reduced-form findings.

## L.8 Validation Against Survey Data

We compare the AI predictions to actual survey data on two observable outcomes—delegation authority and non-performing loan ratios. The AI was never given branch-specific financial data; validation therefore focuses on directional patterns rather than level predictions. For each test, we match branch-years to

the AI's prediction for the actually observed manager ethnicity under the actual conflict level, yielding a one-to-one comparison. Table L3 reports the results.

**Within-branch predictive power.** Regressions of actual outcomes on AI predictions with branch and year fixed effects yield positive but imprecisely estimated coefficients (delegation:  $\hat{\beta} = 5.9$ , s.e. = 5.5; NPL:  $\hat{\beta} = 0.31$ , s.e. = 0.47). The point estimates are consistent with AI predictions moving in the same direction as actual outcomes, but are not statistically distinguishable from zero. This is expected: the AI receives bank-level information but no branch-specific financial data, so it captures average directional effects rather than branch-level outcomes. The remaining tests focus on these directional patterns.

**Delegation direction.** In both 2018 and 2022, the AI and the actual data agree that city-coethnic managers receive *lower* delegation thresholds than bank-coethnic managers (Table L3, Panel B). In the regression specification of Equation (a42), the AI yields  $\hat{\beta}_1 = -158.3$  (s.e. = 30.3) and the full-sample actual data yields  $\hat{\beta}_1 = -1,518.3$  (s.e. = 751.9,  $p < 0.05$ )—both negative (Panel C). The actual-data regression uses the paper's main specification on the full panel ( $N = 644$ ), with inflation-adjusted delegation winsorized at the 95th percentile.

**Coethnic placebo.** The sign reversal on coethnic branches is the cleanest evidence that the misaligned-city results reflect the information-incentive mechanism rather than a generic model bias. We re-run the AI survey on branches in *coethnic* cities, where city-coethnic and bank-coethnic coincide and no information-incentive trade-off exists. The delegation coefficient flips sign:  $\hat{\beta}_1 = +877.1$  (s.e. = 59.0) in coethnic cities versus  $-158.3$  in misaligned cities (Panel C). The information-communication gap flips in the same way, from  $-24.6$  to  $+24.9$  (Panel D). When the trade-off disappears, the AI grants city-coethnic managers more authority, not less.

**NPL mismatch.** The AI predicts higher NPL for city-coethnic managers ( $\hat{\beta}_1 = +0.77$ , s.e. = 0.12), while the actual data produce a negative but imprecisely estimated coefficient ( $\hat{\beta}_1 = -0.34$ , s.e. = 2.19,  $N = 650$ ; Panel C, scaled  $\times 100$ ). In coethnic cities, the AI predicts *lower* NPL for city-coethnic managers ( $\hat{\beta}_1 = -3.22$ ), so its positive prediction in misaligned cities is not a generic bias but a response to the structure of the trade-off. The sign disagreement likely reflects the model weighting the moral-hazard channel more heavily than the screening channel; we cannot distinguish these interpretations from the experiment alone.

**Ethnicity choice.** Unlike the organizational design outcomes, the AI's preferred manager ethnicity does not respond to conflict. The AI chooses a city-coethnic manager in 87.9% of cases regardless of conflict treatment ( $\hat{\beta} = 0.008$ , s.e. = 0.018 on the high-conflict indicator with branch and year fixed effects). In the actual data, conflict shifts manager assignment. The AI captures the organizational design response to conflict (delegation, lending) but not the assignment response, consistent with it overweighting the information advantage of city-coethnic candidates relative to the alignment concerns that drive real personnel choices.

## L.9 Robustness Checks

We assess the robustness of the AI survey results along several dimensions suggested by the literature on LLMs as simulated economic agents (Argyle et al., 2023; Brand, Israeli and Ngwe, 2023; Horton, Filippas and Manning, 2023).

### L.9.1 Model Sensitivity

Our main results use GPT-5-mini, a reasoning model that generates an internal chain of thought before producing answers. To check whether the findings depend on this architectural choice, we replicated the exercise using GPT-4o-mini, a non-reasoning model, with the prompt and experimental design held fixed.<sup>3</sup> The non-reasoning model captures the first-order pattern: city-coethnic managers still score substantially higher on information acquisition (+45 points in GPT-4o-mini versus +55 in GPT-5-mini). It fails, however, on the variables that require joint reasoning about information and alignment. Information communication flips from −23 to +8, ethnic bias reverses from +40 to −4, delegation reverses from −158 to +233 thousand ETB, and expected NPL flips from +0.8 to −3.0 percentage points.

This is the same failure mode that the *minimal* prompt induces in GPT-5-mini (Section L.9.2): positive delegation, positive information communication, negligible or negative ethnic bias, and lower expected NPL. Whether one strips the agency structure from the *prompt* or uses a *model* that does not integrate information and alignment in the same judgment, the output defaults to a “local knowledge is uniformly good” heuristic—recognizing the information advantage of city-coethnic managers but not connecting it to moral hazard or tighter headquarters control. We therefore use GPT-5-mini for all reported results.

### L.9.2 Prompt Sensitivity

A natural concern with AI survey experiments is that the model’s responses may reflect the prompt rather than its latent knowledge. Our main specification uses a “subtle” prompt that provides the CEO with factual context about Ethiopian banking, including a description of the branch lending process: how managers acquire information about borrowers, communicate recommendations to headquarters, and make autonomous decisions below a delegation threshold. This context also describes how managers develop accountability relationships with local communities and with headquarters, noting that local ties create social obligations while headquarters ties create career accountability. While this framing avoids explicit mention of trade-offs, a skeptical reader might argue that describing these mechanisms primes the model to reproduce them.

To address this concern, we replicate the experiment using a “naive” prompt that provides only minimal context: ethnicity is described as “a demographic characteristic that varies across regions,” and banks are described as serving “customers across different regions with varying ethnic compositions.” Critically, the naive prompt retains the full lending process context, so any differences reflect the ethnicity framing rather than the institutional description.

Table L4 reports the results. The core findings are remarkably stable across prompt versions. Panel A shows that city-coethnic managers score 53 points higher on information acquisition under the naive prompt

<sup>3</sup> The GPT-4o-mini replication used the *subtle* prompt on 98 of the 100 misaligned branches—196 of 200 branch-years—with the remaining 2 branches lost to API failures.

versus 55 under the subtle prompt, 21 points lower on information communication (versus –23), and 38 points higher on ethnic bias (versus 40). The information-alignment trade-off emerges with nearly identical magnitudes regardless of how ethnicity is framed.

Panel B confirms that conflict treatment effects are also robust. High conflict reduces delegation thresholds and compresses lending volumes under both prompts. The conflict  $\times$  city-coethnic interaction on lending volume—our key measure of how conflict differentially affects expectations about city-coethnic versus bank-coethnic managers—is –14 million ETB under the naive prompt and –15 under the subtle prompt. The delegation level effect is also robust: city-coethnic managers receive –214 thousand ETB lower delegation under the naive prompt versus –158 under the subtle prompt, both highly significant ( $p < 0.01$ ). The naive estimate is somewhat larger in magnitude, indicating that the minimal ethnicity context does not weaken the model’s ability to translate the information-incentive trade-off into organizational design responses.

**The role of institutional context.** Both the subtle and naive prompts include a description of the branch lending process: how managers gather information about borrowers, communicate recommendations to headquarters, make autonomous decisions below a delegation threshold, and develop accountability relationships with local communities and with bank headquarters. A skeptical reader might argue that this description primes the model to reproduce the information-incentive trade-off. To test this, we run a “minimal” prompt that strips all mechanism framing from the lending process description, retaining only the bare logistical steps (assess applicants, recommend to headquarters, decide autonomously on small loans) without any mention of soft information, verification problems, favoritism pressures, or accountability relationships.

Table L5 shows that the institutional context is doing essential work. Without the lending-process description, the information-acquisition advantage for city-coethnic managers survives but attenuates from +55 to +38 points: the model still recognizes their local-knowledge advantage. Three other variables, however, reverse sign. Information communication flips from –23 to +20; without the non-verifiability of soft information, the model treats local knowledge as equally useful for gathering and for transmitting. Ethnic bias flips from +40 to –4; without community pressure in the prompt, the model sees no reason to associate local embeddedness with biased lending. And delegation flips from –158 to +346 thousand ETB; the model interprets local knowledge as a competence signal that should be rewarded with greater autonomy.

The lending and NPL predictions under the minimal prompt are internally implausible. City-coethnic managers are expected to generate substantially more lending (+60 million ETB versus +44 under the subtle prompt) while simultaneously producing *lower* NPL rates (–3.0 percentage points versus +0.8). More lending with fewer defaults implies that the model sees no cost to the local information advantage. Under the full institutional context, the NPL sign reverses to marginally positive, reflecting that local ties create incentives to approve marginal loans.

The delegation comparison to the actual data is the punchline. The minimal prompt predicts that city-coethnic managers should receive *more* delegation (+346 thousand ETB), which directly contradicts the empirical estimate of –1,518 thousand ETB (Table L3 Panel C). The full-context estimate (–158) recovers the correct sign. The stripped-context model gets the direction of organizational design wrong because it lacks the economic structure needed to translate an information advantage into a moral-hazard concern. That the full-context results align with the observed data while the stripped-context results do not is consistent

with the agency-theoretic interpretation of the organizational patterns we document, rather than with a simple competence story.

### **L.9.3 Other Design Considerations**

**Temperature sensitivity.** GPT-5-mini only accepts temperature 1.0, so the standard temperature-sensitivity check is not available for this model. The model-sensitivity analysis above partially substitutes by comparing architectures with different stochastic generation processes.

**Question-order effects.** The survey places perception scores and business outcomes first (Q1–Q14), mechanism weights second (Q15), and the manager ethnicity choice last (Q16–Q17). This order ensures that the final choice synthesizes prior analysis rather than anchoring subsequent scores.

**Within-scenario variance.** We do not run each scenario multiple times. Re-running each scenario  $k$  times would separate within-scenario measurement noise from systematic cross-scenario differences. The current design relies instead on cross-branch variation for statistical power. This is a limitation rather than a design choice: budget constraints preclude full  $k$ -fold replication at the present scale, and we note it as an avenue for extension.

Table L3: External Validation: AI Survey vs Actual Data

| <i>Panel A: Manager Ethnicity Choice</i>                                        |                      |                       |                       |
|---------------------------------------------------------------------------------|----------------------|-----------------------|-----------------------|
|                                                                                 | AI Survey            | Actual Data           | Coethnic Placebo      |
| Share choosing city-coethnic                                                    | 87.9%                | 33.7%                 | 100.0%                |
| Share choosing bank-coethnic                                                    | 6.3%                 | 53.7%                 | —                     |
| AI recommendation = actual hire (%)                                             |                      | 33.7%                 | 69.4%                 |
| Random baseline                                                                 |                      | 25%                   | 25%                   |
| N (branch-years)                                                                |                      | 190                   | 98                    |
| <i>Panel B: City-Coethnic Effect (Matched Data, z-Score Gap)</i>                |                      |                       |                       |
|                                                                                 | AI                   | Actual                | Coethnic Placebo (AI) |
| <i>Delegation gap (city – comparison):</i>                                      |                      |                       |                       |
| 2018                                                                            | -0.014               | -0.256                | 1.172                 |
| 2022                                                                            | -0.553               | -0.450                | 1.373                 |
| <i>NPL gap (city – comparison):</i>                                             |                      |                       |                       |
| 2018                                                                            | 0.264                | -0.386                | -0.630                |
| 2022                                                                            | 0.422                | -0.014                | -1.223                |
| <i>Panel C: Regression Coefficients (<math>\beta_1</math> on City-Coethnic)</i> |                      |                       |                       |
|                                                                                 | AI (Misaligned)      | Actual (Misaligned)   | AI (Coethnic)         |
| Delegation                                                                      | -158.3***<br>( 30.3) | -1518.3**<br>( 751.9) | 877.1***<br>( 59.0)   |
| NPL ( $\times 100$ )                                                            | 0.77***<br>( 0.12)   | -0.34<br>( 2.19)      | -3.22***<br>( 0.11)   |
| <i>High conflict <math>\times</math> city-coethnic (<math>\beta_3</math>):</i>  |                      |                       |                       |
| Delegation                                                                      | -2.0<br>( 48.8)      | —                     | —                     |
| NPL                                                                             | 0.39*<br>( 0.21)     | —                     | —                     |
| N (delegation)                                                                  | 742                  | 644                   | 784                   |
| N (NPL)                                                                         | 742                  | 650                   | 784                   |
| Branch FE                                                                       | Yes                  | Yes                   | Yes                   |
| Year FE                                                                         | Yes                  | Yes                   | Yes                   |
| <i>Panel D: Internal Mechanism Consistency (AI Only, Misaligned)</i>            |                      |                       |                       |
|                                                                                 | Gap (city – bank)    | Banks unanimous       | Placebo gap           |
| Info acquisition                                                                | 54.7<br>[ 5.9]       | 13/13 positive        | 52.7***<br>( 0.6)     |
| Info communication                                                              | -24.6<br>[ 8.9]      | 13/13 negative        | 24.9***<br>( 0.5)     |
| Trade-off (both conditions)                                                     |                      | 13/13 banks           |                       |
| Delegation                                                                      | 4.4                  | 5/13 positive         | 877.1***              |
| NPL                                                                             | 0.38                 | 7/13 positive         | -3.22***              |

*Notes:* Panels A–B compare AI predictions and actual survey data for a matched sample of 150 Ethiopian bank branches from the AI survey subsample. Panel C reports regression coefficients from the paper’s main specification: AI regressions use the experimental design with demeaned high-conflict treatment and branch/year FE; actual-data regressions use the *full panel* (not the AI subsample) with branch and year FE. Delegation is inflation-adjusted (deflated to 2018 ETB) and winsorized at the 95th percentile within period  $\times$  bank-city coethnicity cells, matching the paper’s main specification. NPL in the actual data is in share form (0–1); coefficients are scaled  $\times 100$  for comparability with the AI’s percentage-point scale. Columns labeled “Misaligned” restrict to cities where bank and city have different dominant ethnicities (the paper’s main sample); “Coethnic” restricts to cities where bank and city share ethnicity. Standard errors clustered at the branch level in parentheses. Panel D reports the AI’s internal mechanism scores: the gap between city-coethnic and bank-coethnic managers on information acquisition (0–100) and communication (0–100), with bank-level unanimity counts. Standard deviations in brackets; standard errors from coethnic placebo regressions in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table L4: AI Survey: Prompt Robustness (Naive Prompt)

| <i>Panel A: CEO Assessments and Mechanism Weights</i> |                       |                     |                    |                     |                    |                    |                    |                    |
|-------------------------------------------------------|-----------------------|---------------------|--------------------|---------------------|--------------------|--------------------|--------------------|--------------------|
|                                                       | (1)                   | (2)                 | (3)                | (4)                 | (5)                | (6)                | (7)                | (8)                |
|                                                       | CEO Assessments       |                     |                    |                     | Mechanism Weights  |                    |                    |                    |
|                                                       | Info Acq.             | Info Comm.          | Ethnic Bias        | Safety Concern      | Info               | Align              | Safety             | Other              |
| City-coethnic                                         | 53.37***<br>(0.48)    | -21.12***<br>(0.45) | 38.37***<br>(0.80) | -28.36***<br>(0.87) | 12.75***<br>(0.31) | -7.04***<br>(0.31) | -9.28***<br>(0.35) | 3.57***<br>(0.26)  |
| High conflict                                         | -5.74***<br>(0.46)    | -0.89**<br>(0.41)   | 5.32***<br>(1.26)  | 31.87***<br>(1.12)  | -4.07***<br>(0.50) | -4.18***<br>(0.45) | 9.48***<br>(0.53)  | -1.23***<br>(0.26) |
| High conflict × City-coethnic                         | 4.71***<br>(0.52)     | -4.27***<br>(0.80)  | -2.50*<br>(1.34)   | -9.77***<br>(1.61)  | 0.97<br>(0.64)     | 3.76***<br>(0.56)  | -3.37***<br>(0.61) | -1.36***<br>(0.41) |
| Branch FE                                             | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                | Yes                |
| Year FE                                               | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                | Yes                |
| Dep. var. mean                                        | 57.59                 | 70.14               | 51.55              | 44.76               | 35.35              | 36.71              | 18.29              | 9.66               |
| Observations                                          | 742                   | 742                 | 739                | 742                 | 742                | 742                | 742                | 742                |
| R <sup>2</sup>                                        | 0.978                 | 0.821               | 0.829              | 0.803               | 0.747              | 0.565              | 0.701              | 0.389              |
| <i>Panel B: Organizational Outcomes</i>               |                       |                     |                    |                     |                    |                    |                    |                    |
|                                                       | (1)                   | (2)                 | (3)                | (4)                 | (5)                | (6)                | (7)                | (8)                |
|                                                       | Outcomes              |                     |                    |                     | Lending (%)        |                    | NPL (%)            |                    |
|                                                       | Deleg.                | Wage Prem.          | Exp. NPL           | Lend Vol.           | City-Coeth.        | Bank-Coeth.        | City-Coeth.        | Bank-Coeth.        |
| City-coethnic                                         | -213.54***<br>(38.60) | -25.33***<br>(0.56) | 0.74***<br>(0.11)  | 43.01***<br>(1.76)  | 0.29<br>(0.40)     | -5.40***<br>(0.40) | 1.71***<br>(0.16)  | -2.32***<br>(0.22) |
| High conflict                                         | -237.89***<br>(50.65) | 9.92***<br>(0.43)   | 2.19***<br>(0.17)  | -16.21***<br>(1.85) | -0.59<br>(0.59)    | 0.95*<br>(0.52)    | 2.04***<br>(0.20)  | 2.84***<br>(0.34)  |
| High conflict × City-coethnic                         | 48.12<br>(60.26)      | -2.27**<br>(0.99)   | 0.55**<br>(0.25)   | -14.09***<br>(2.83) | 1.89**<br>(0.79)   | -2.16***<br>(0.63) | 1.31***<br>(0.31)  | -1.58***<br>(0.50) |
| Branch FE                                             | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                | Yes                |
| Year FE                                               | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                | Yes                |
| Dep. var. mean                                        | 783.63                | 10.18               | 7.64               | 97.74               | 66.06              | 17.73              | 8.13               | 7.02               |
| Observations                                          | 742                   | 742                 | 742                | 742                 | 742                | 742                | 742                | 742                |
| R <sup>2</sup>                                        | 0.449                 | 0.843               | 0.605              | 0.740               | 0.533              | 0.530              | 0.567              | 0.402              |

*Notes:* This table replicates Table 7 using a “naive” prompt that provides minimal context about the role of ethnicity in Ethiopian banking. The prompt describes ethnicity as “a demographic characteristic” without mentioning information networks, trust, or incentive trade-offs. The lending process context (information acquisition, communication to headquarters, autonomous decisions, and manager accountability relationships) is identical to the main specification. All other aspects of the experimental design—branch sample, conflict treatments, manager types, and survey questions—are unchanged. Sample: 98 misaligned branches (of 100 targeted; 2 branches lost due to API failures). Standard errors clustered at branch level in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

Table L5: AI Survey: Institutional Context Ablation (Minimal Prompt)

| <i>Panel A: CEO Assessments and Mechanism Weights</i> |                       |                     |                    |                     |                    |                    |                     |                    |
|-------------------------------------------------------|-----------------------|---------------------|--------------------|---------------------|--------------------|--------------------|---------------------|--------------------|
|                                                       | (1)                   | (2)                 | (3)                | (4)                 | (5)                | (6)                | (7)                 | (8)                |
|                                                       | CEO Assessments       |                     |                    |                     | Mechanism Weights  |                    |                     |                    |
|                                                       | Info Acq.             | Info Comm.          | Ethnic Bias        | Safety Concern      | Info               | Align              | Safety              | Other              |
| City-coethnic                                         | 38.30***<br>(0.69)    | 20.09***<br>(0.64)  | -4.01***<br>(1.00) | -29.10***<br>(0.84) | 8.98***<br>(0.30)  | 2.38***<br>(0.36)  | -12.33***<br>(0.33) | 0.98***<br>(0.23)  |
| High conflict                                         | -11.36***<br>(0.91)   | -11.08***<br>(1.02) | 13.42***<br>(1.40) | 27.21***<br>(1.08)  | -5.11***<br>(0.50) | -6.74***<br>(0.50) | 12.00***<br>(0.56)  | -0.15<br>(0.29)    |
| High conflict × City-coethnic                         | 5.86***<br>(1.14)     | 6.11***<br>(1.19)   | -3.23*<br>(1.81)   | -3.15*<br>(1.83)    | 2.15***<br>(0.68)  | 5.27***<br>(0.53)  | -5.54***<br>(0.63)  | -1.89***<br>(0.40) |
| Branch FE                                             | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                 | Yes                |
| Year FE                                               | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                 | Yes                |
| Dep. var. mean                                        | 60.60                 | 67.81               | 38.88              | 47.24               | 35.80              | 34.43              | 20.90               | 8.87               |
| Observations                                          | 741                   | 742                 | 727                | 742                 | 742                | 742                | 742                 | 742                |
| R <sup>2</sup>                                        | 0.876                 | 0.679               | 0.337              | 0.783               | 0.642              | 0.401              | 0.770               | 0.190              |
| <i>Panel B: Organizational Outcomes</i>               |                       |                     |                    |                     |                    |                    |                     |                    |
|                                                       | (1)                   | (2)                 | (3)                | (4)                 | (5)                | (6)                | (7)                 | (8)                |
|                                                       | Outcomes              |                     |                    |                     | Lending (%)        |                    | NPL (%)             |                    |
|                                                       | Deleg.                | Wage Prem.          | Exp. NPL           | Lend Vol.           | City-Coeth.        | Bank-Coeth.        | City-Coeth.         | Bank-Coeth.        |
| City-coethnic                                         | 345.55***<br>(33.04)  | -22.24***<br>(0.44) | -2.98***<br>(0.14) | 60.04***<br>(2.13)  | 5.35***<br>(0.48)  | -7.17***<br>(0.49) | -2.75***<br>(0.14)  | -4.67***<br>(0.27) |
| High conflict                                         | -286.25***<br>(37.16) | 9.21***<br>(0.39)   | 3.35***<br>(0.20)  | -26.32***<br>(2.36) | -4.08***<br>(0.73) | 4.06***<br>(0.57)  | 2.26***<br>(0.19)   | 6.37***<br>(0.39)  |
| High conflict × City-coethnic                         | 40.95<br>(54.42)      | 0.93<br>(0.98)      | -0.80***<br>(0.21) | -8.63**<br>(3.57)   | 2.78***<br>(0.86)  | -2.81***<br>(0.69) | 0.28<br>(0.23)      | -3.31***<br>(0.40) |
| Branch FE                                             | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                 | Yes                |
| Year FE                                               | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                 | Yes                |
| Dep. var. mean                                        | 861.40                | 12.74               | 6.66               | 104.14              | 65.18              | 20.43              | 6.27                | 7.78               |
| Observations                                          | 741                   | 742                 | 741                | 742                 | 742                | 740                | 739                 | 742                |
| R <sup>2</sup>                                        | 0.629                 | 0.821               | 0.700              | 0.785               | 0.609              | 0.572              | 0.665               | 0.617              |

*Notes:* This table replicates Table 7 using a “minimal” prompt that strips all mechanism framing from the lending process description. The minimal prompt retains only the bare logistical steps (manager assesses applicants, recommends to headquarters, decides autonomously on small loans) without any mention of soft information, verification problems, favoritism pressures, or accountability relationships. The ethnicity context is identical to the naive prompt (Table L4). Sign reversals on information communication, ethnic bias, delegation, and NPL indicate that the institutional context is necessary for the model to reason about the information-incentive trade-off rather than defaulting to a naive “local knowledge is always beneficial” heuristic. Sample: 98 misaligned branches. Standard errors clustered at branch level in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table L6: AI Survey: Mechanism Weights and Lending by Manager Ethnicity

| <i>Panel A: Manager Type Differences (Actual Conflict)</i> |                       |                     |                    |                     |                    |                    |                    |                    |
|------------------------------------------------------------|-----------------------|---------------------|--------------------|---------------------|--------------------|--------------------|--------------------|--------------------|
|                                                            | (1)                   | (2)                 | (3)                | (4)                 | (5)                | (6)                | (7)                | (8)                |
|                                                            | CEO Assessments       |                     |                    |                     | Mechanism Weights  |                    |                    |                    |
|                                                            | Info Acq.             | Info Comm.          | Ethnic Bias        | Safety Concern      | Info               | Align              | Safety             | Other              |
| City-coethnic                                              | 54.52***<br>(0.35)    | -22.83***<br>(0.47) | 39.87***<br>(0.68) | -27.80***<br>(0.81) | 12.47***<br>(0.31) | -7.01***<br>(0.33) | -9.19***<br>(0.30) | 3.73***<br>(0.24)  |
| City-coethnic ×<br>Year 2022                               | 0.77<br>(0.48)        | 0.31<br>(0.89)      | -1.07<br>(1.04)    | 1.76<br>(1.38)      | 0.38<br>(0.63)     | -0.31<br>(0.61)    | 0.89<br>(0.54)     | -0.95**<br>(0.47)  |
| Branch FE                                                  | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                | Yes                |
| Year FE                                                    | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                | Yes                |
| Dep. var. mean                                             | 58.04                 | 69.28               | 51.49              | 44.49               | 34.77              | 37.43              | 18.45              | 9.35               |
| Observations                                               | 742                   | 742                 | 742                | 742                 | 742                | 742                | 742                | 742                |
| R <sup>2</sup>                                             | 0.982                 | 0.810               | 0.874              | 0.460               | 0.685              | 0.484              | 0.460              | 0.320              |
| <i>Panel B: Effect of Conflict (High vs Low)</i>           |                       |                     |                    |                     |                    |                    |                    |                    |
|                                                            | (1)                   | (2)                 | (3)                | (4)                 | (5)                | (6)                | (7)                | (8)                |
|                                                            | Outcomes              |                     |                    |                     | Lending (%)        |                    | NPL (%)            |                    |
|                                                            | Deleg.                | Wage Prem.          | Exp. NPL           | Lend Vol.           | City-Coeth.        | Bank-Coeth.        | City-Coeth.        | Bank-Coeth.        |
| City-coethnic                                              | -158.21***<br>(30.21) | -25.73***<br>(0.50) | 0.77***<br>(0.12)  | 44.30***<br>(1.83)  | 1.26***<br>(0.46)  | -5.13***<br>(0.42) | 1.91***<br>(0.16)  | -2.69***<br>(0.25) |
| City-coethnic ×<br>Year 2022                               | 76.93<br>(57.87)      | 0.15<br>(0.84)      | -0.24<br>(0.21)    | 4.63*<br>(2.72)     | 2.07***<br>(0.67)  | -1.22*<br>(0.67)   | -0.46<br>(0.28)    | 0.13<br>(0.43)     |
| Branch FE                                                  | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                | Yes                |
| Year FE                                                    | Yes                   | Yes                 | Yes                | Yes                 | Yes                | Yes                | Yes                | Yes                |
| Dep. var. mean                                             | 704.85                | 9.73                | 7.77               | 94.08               | 65.88              | 17.80              | 8.38               | 6.83               |
| Observations                                               | 742                   | 742                 | 742                | 742                 | 742                | 742                | 742                | 742                |
| R <sup>2</sup>                                             | 0.410                 | 0.778               | 0.386              | 0.677               | 0.499              | 0.501              | 0.420              | 0.305              |

*Notes:* This table reports the full set of AI-survey outcomes, extending Table 7. Panel A shows CEO assessments (columns 1–4) and the mechanism-weight allocation across information, alignment, safety, and other (columns 5–8). Panel B shows organizational outcomes (columns 1–4) and lending and NPL shares by customer ethnicity (columns 5–8). Regressors are an indicator for a city-coethnic manager and its interaction with a 2022-wave indicator, absorbing branch and year fixed effects. Sample restricted to branch-years in non-coethnic cities. Standard errors clustered at branch level in parentheses. Statistical significance is denoted by \*  $p < 0.10$ , \*\*  $p < 0.05$ , and \*\*\*  $p < 0.01$ .

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